IT Operations Scenarios: process, flowchart and proof

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How I would run each common IT operations need, step by step, with evidence from my real work and my labs. Lab work uses synthetic data.

IT operations scenarios

These are 16 common IT operations scenarios. Each page shows four things:

- what I did in real work at TikTok (IT Operations Engineer, Dublin, EMEA, March 2023 to September 2026)
- the process I follow, as a flowchart and numbered steps
- the proof in this repository: screenshots, build log steps, manuals, scripts and tests
- the tools, and where a different tool runs the same process

When I did not own a task in real work, the page says so.

#	Scenario	Link
01	Jira Service Management admin	Open · page 2
02	New starter process	Open · page 6
03	Service desk backlog and routing	Open · page 9
04	Mac and Windows device management	Open · page 14
05	Non-compliant Mac loses Microsoft 365 access	Open · page 18
06	External collaboration in Microsoft 365	Open · page 19
07	Joiner, mover and leaver automation	Open · page 22
08	Office internet outage: the first 60 minutes	Open · page 24
09	Shared file service recovery	Open · page 26
10	Missing patches and open security alerts	Open · page 27
11	SaaS licence and external access request	Open · page 29
12	Developer cannot access a critical service	Open · page 32
13	Systems inventory and ownership	Open · page 33
14	On-site prioritisation	Open · page 35
15	Process improvement: imaging from 60 to 15 minutes	Open · page 38
16	Firewall management	Open · page 40

Most screenshots come from my Jira Service Management lab in Project 07. The build log shows how I built it, step by step.

Lab work uses synthetic data. No employer data is published.

Jira Service Management admin

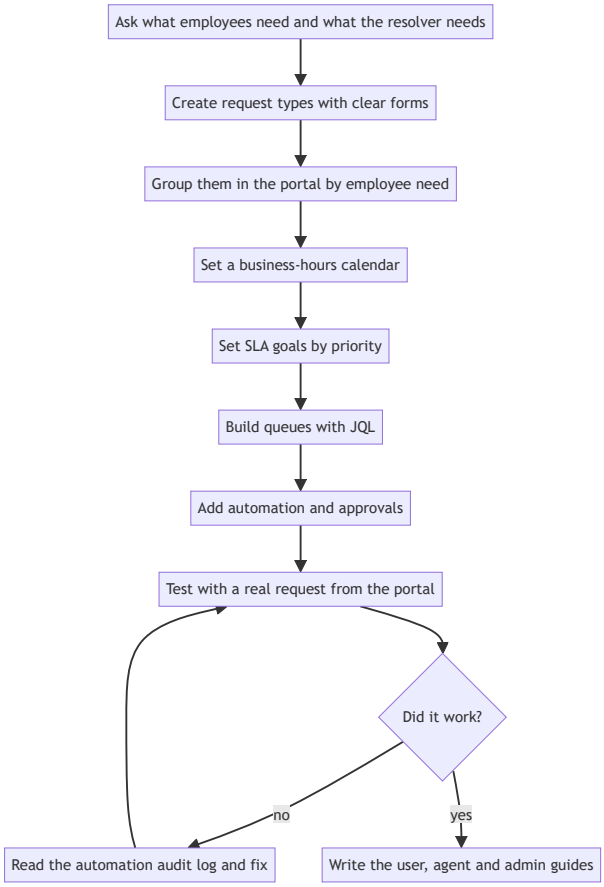
The situation: A company runs its IT service desk on Jira Service Management. It needs someone who can change the setup: request types, SLAs, queues, automation and approvals.

What I did in real work

At TikTok I used Jira every day as an agent, not as an admin. I set priorities, owned L2 and L3 escalations, worked to SLAs and wrote runbooks in Lark.

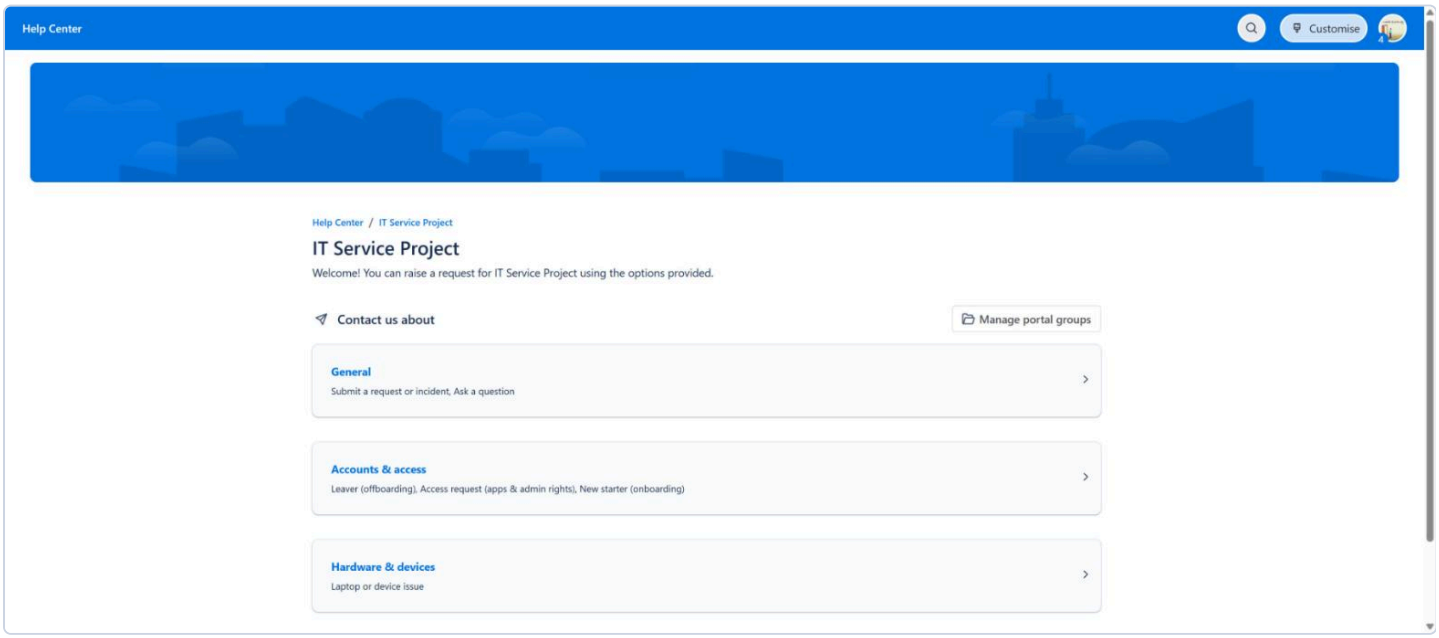
To learn the admin side, I built my own Jira Service Management site and configured it from the start. The proof on this page comes from that lab site.

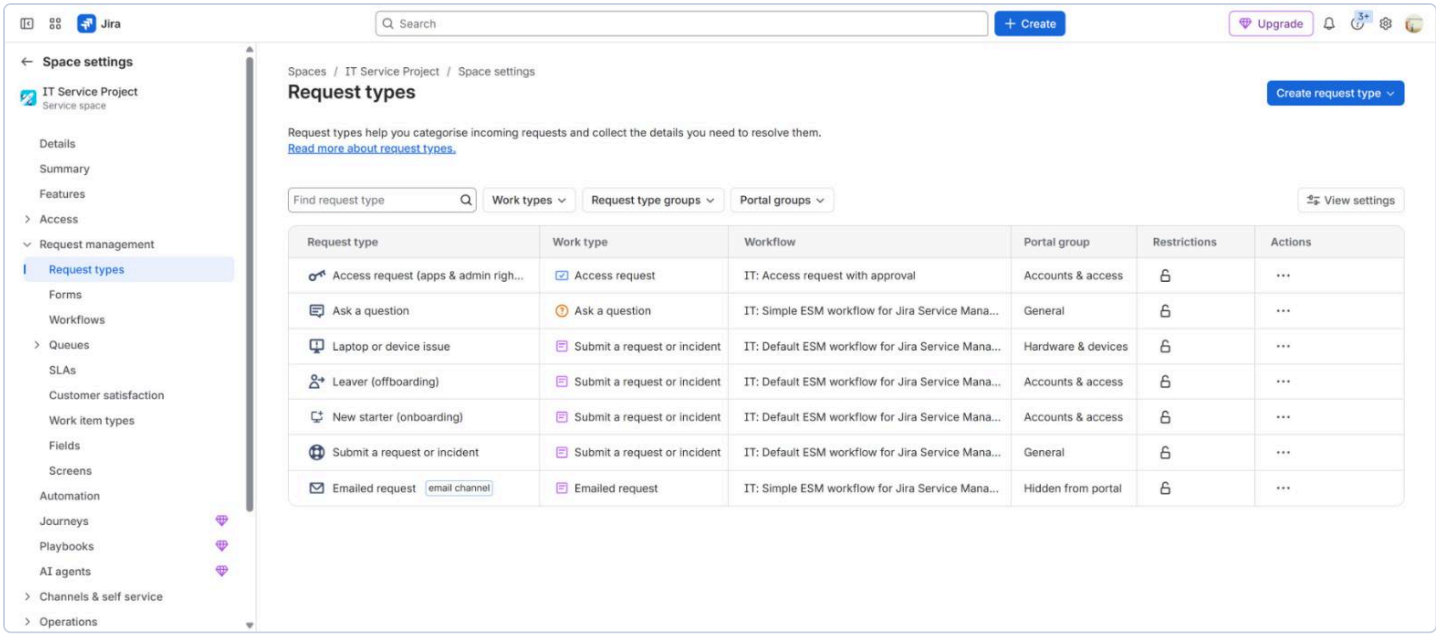
The process I follow

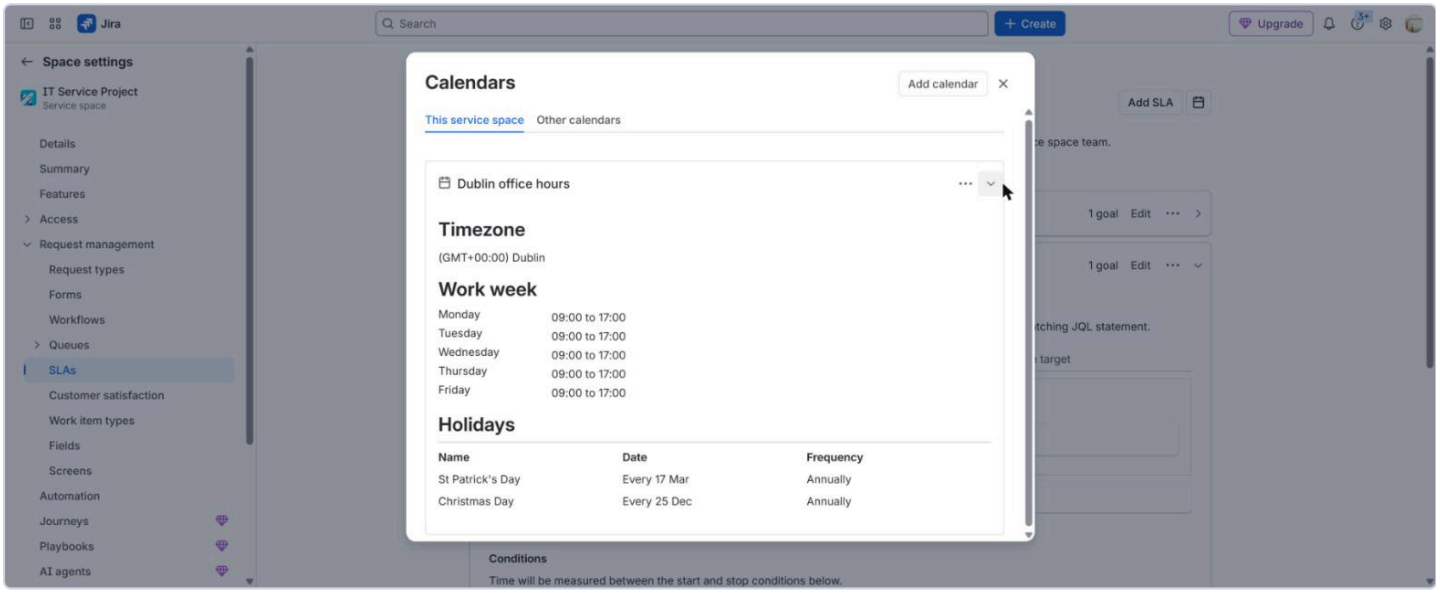


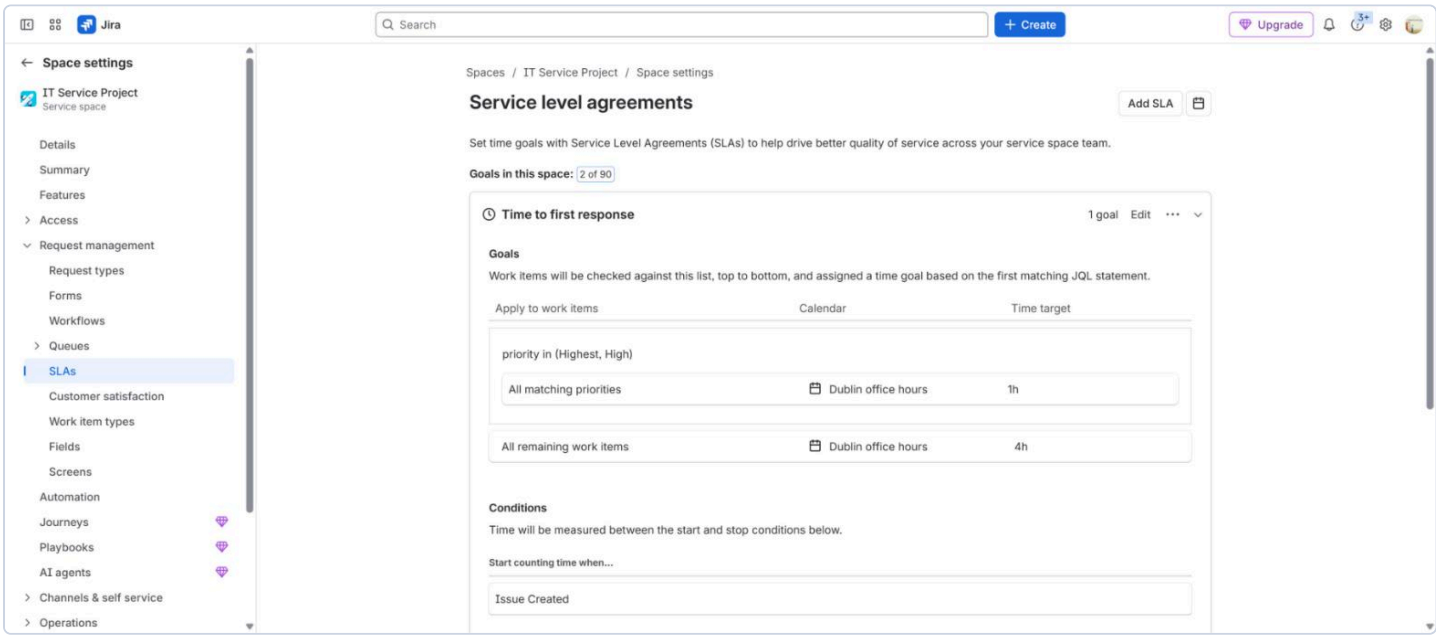
1. Ask what employees need and what the person who resolves the ticket needs to know.
2. Create one request type per need, with a form that asks for those details.
3. Group the request types in the portal by what the employee needs, not by how IT sorts work.
4. Set a calendar with the real office hours and public holidays.
5. Set SLA goals by priority, with a catch-all goal last.
6. Build queues with JQL so agents see the most urgent work first.
7. Add automation for repeat steps and approvals for access.
8. Raise a real test request through the portal.
9. If the result is wrong, open the automation audit log, find the failed step and fix it.
10. Write short guides for users, agents and admins.

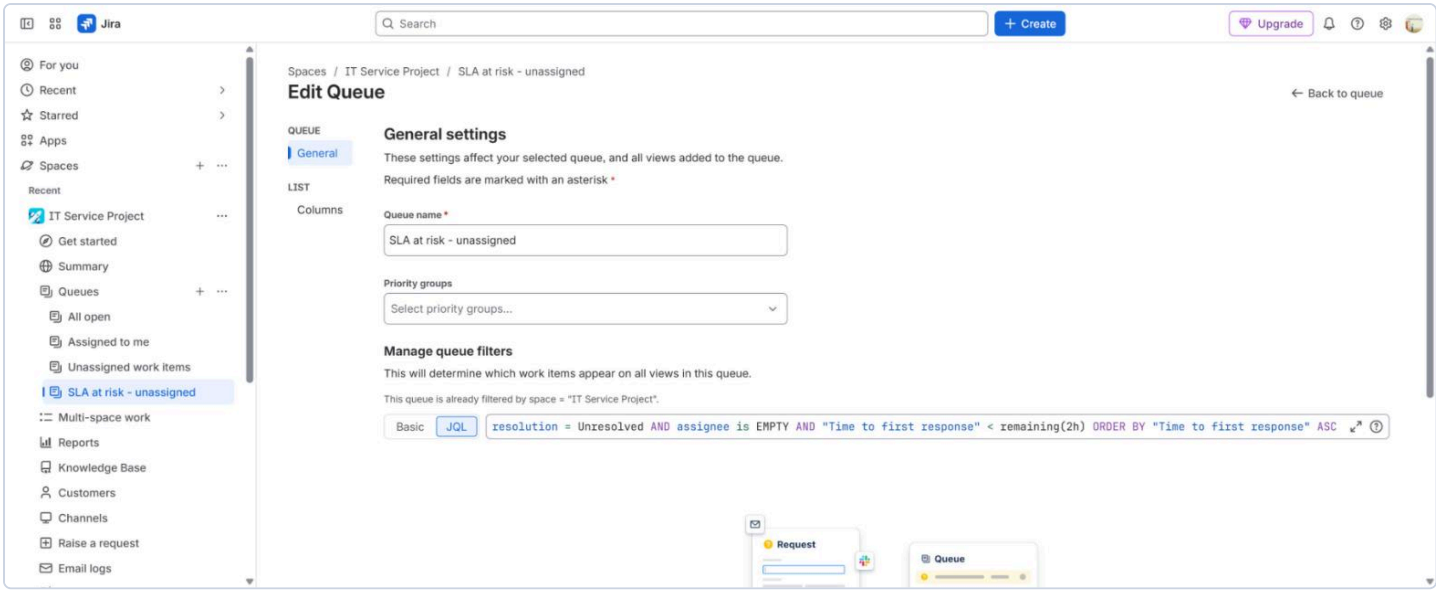
Proof

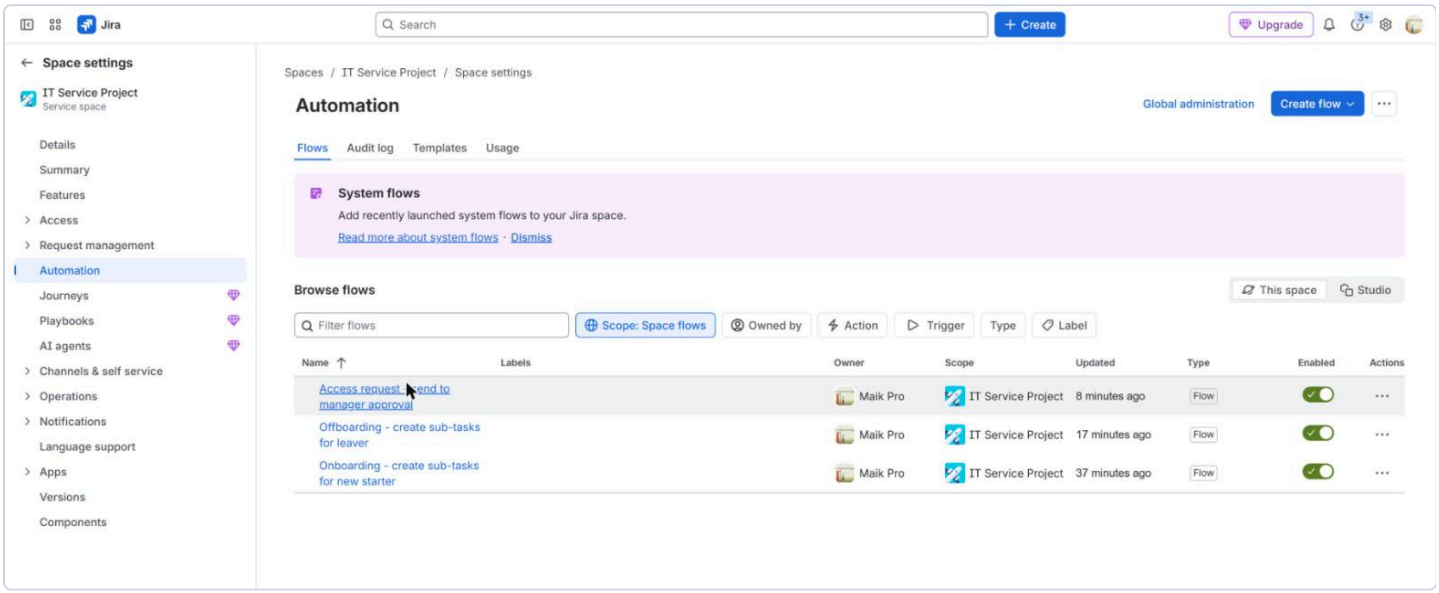


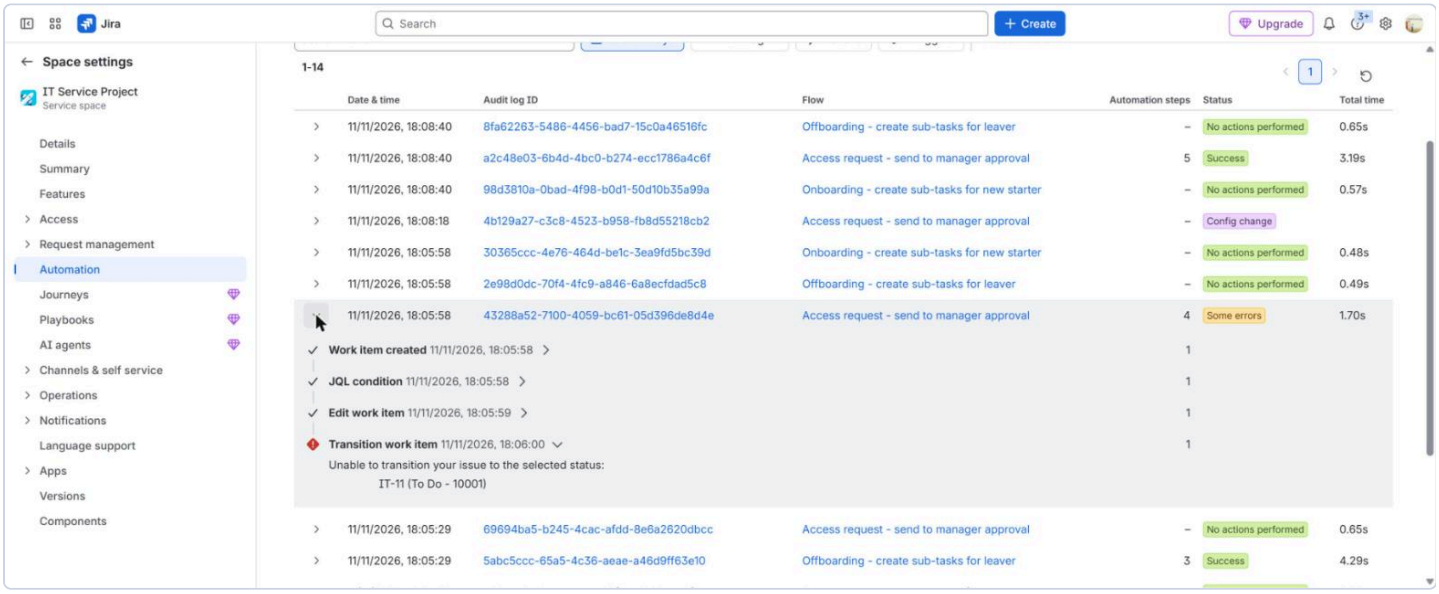

Jira
Search
Create
Upgrade
Space settings
IT Service Project
Service space
Details
Summary
Features
Access
Request management
Request types
Forms
Workflows
Queues
SLAs
Customer satisfaction
Work item types
Fields
Screens
Automation
Journeys
Playbooks
AI agents
Channels & self service
Operations
Spaces / IT Service Project / Space settings
Request types
Create request type
Request types help you categorise incoming requests and collect the details you need to resolve them.
Read more about request types.
Find request type
Work types
Request type groups
Portal groups
View settings
Request type
Work type
Workflow
Portal group
Restrictions
Actions
Access request (apps & admin righ...
Access request
IT: Access request with approval
Accounts & access
Ask a question
Ask a question
IT: Simple ESM workflow for Jira Service Mana...
General
Laptop or device issue
Submit a request or incident
IT: Default ESM workflow for Jira Service Mana...
Hardware & devices
Leaver (offboarding)
Submit a request or incident
IT: Default ESM workflow for Jira Service Mana...
Accounts & access
New starter (onboarding)
Submit a request or incident
IT: Default ESM workflow for Jira Service Mana...
Accounts & access
Submit a request or incident
Submit a request or incident
IT: Default ESM workflow for Jira Service Mana...
General
Emailed request
email channel
Emailed request
IT: Simple ESM workflow for Jira Service Mana...
Hidden from portal


Jira
Search
Create
Upgrade
Space settings
IT Service Project
Service space
Calendars
Add calendar
This service space
Other calendars
Dublin office hours
Timezone
(GMT+00:00) Dublin
Work week
Monday 09:00 to 17:00
Tuesday 09:00 to 17:00
Wednesday 09:00 to 17:00
Thursday 09:00 to 17:00
Friday 09:00 to 17:00
Holidays
Name
Date
Frequency
St Patrick's Day
Every 17 Mar
Annually
Christmas Day
Every 25 Dec
Annually
Add SLA
1 goal
Edit
Conditions
Time will be measured between the start and stop conditions below.


Jira
Search
Create
Upgrade
Space settings
IT Service Project
Service space
Spaces / IT Service Project / Space settings
Service level agreements
Add SLA
Set time goals with Service Level Agreements (SLAs) to help drive better quality of service across your service space team.
Goals in this space: 2 of 90
Time to first response
1 goal
Edit
Goals
Work items will be checked against this list, top to bottom, and assigned a time goal based on the first matching JQL statement.
Apply to work items
Calendar
Time target
priority in (Highest, High)
All matching priorities
Dublin office hours
1h
All remaining work items
Dublin office hours
4h
Conditions
Time will be measured between the start and stop conditions below.
Start counting time when...
Issue Created







- Build log, Steps 1 to 15
- Build log, Step 9: testing and troubleshooting
- Admin guide
- Agent runbook
- User guide
- Phase 5: permission, notification and workflow schemes
- Test results

Tools

- Jira Service Management Cloud, JQL, Jira automation, Assets, Confluence knowledge base
- Not configured in the lab: SAML SSO and SCIM from Entra ID. They are a next step.

Lab work uses synthetic data. No employer data is published.

New starter process

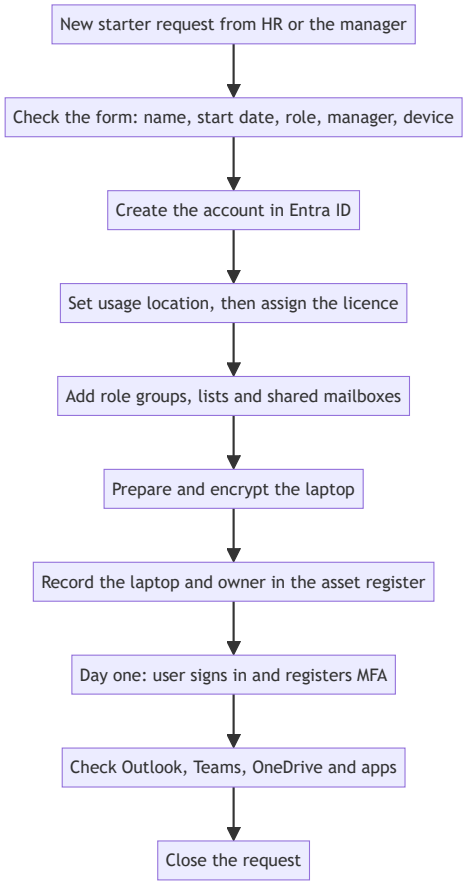
The situation: HR tells IT that a new person starts next Monday. On day one they need an account, the right access and a ready laptop.

What I did in real work

At TikTok I handled joiners in Entra ID and Microsoft 365. I created accounts, reset passwords and MFA, assigned licences, and added people to security groups, distribution lists and shared mailboxes. I prepared Mac laptops with Apple Business Manager and Puppet, and Windows laptops with BitLocker. I applied the laptop image with MCU and checked each machine against a hardware checklist before handover.

I also worked daily in an internal ITAM system, which was the record of truth for equipment. I looked up and updated user records, added new laptops to the inventory, and set the asset to the user at handover. When a device came back I updated the same record, so the inventory matched what people actually held. A joiner or leaver was not finished for me until that record was correct.

The process I follow



1. Receive the New starter request from HR or the manager.
2. Check the form has the name, start date, role, manager and device type.
3. Create the account in Entra ID with the standard name format.
4. Set the usage location.
5. Assign the Microsoft 365 licence.
6. Add the person to the security groups and distribution lists for the role.
7. Give shared mailbox access if the owner approved it.
8. Prepare the laptop. Mac: Apple Business Manager and Puppet. Windows: standard build and BitLocker.
9. Record the laptop and its owner in the asset register. At TikTok this was an internal ITAM system; in my lab it is Jira Assets.
10. On day one, the user signs in and registers MFA.
11. Check Outlook, Teams, OneDrive and the required apps with the user.
12. Close the request.

The asset record: real work and my lab

At TikTok the equipment record lived in an internal ITAM system, and I used it as described above. I could not take that system with me, so in my own lab I rebuilt the same idea in Jira Assets to learn how to configure and administer it rather than only use it.

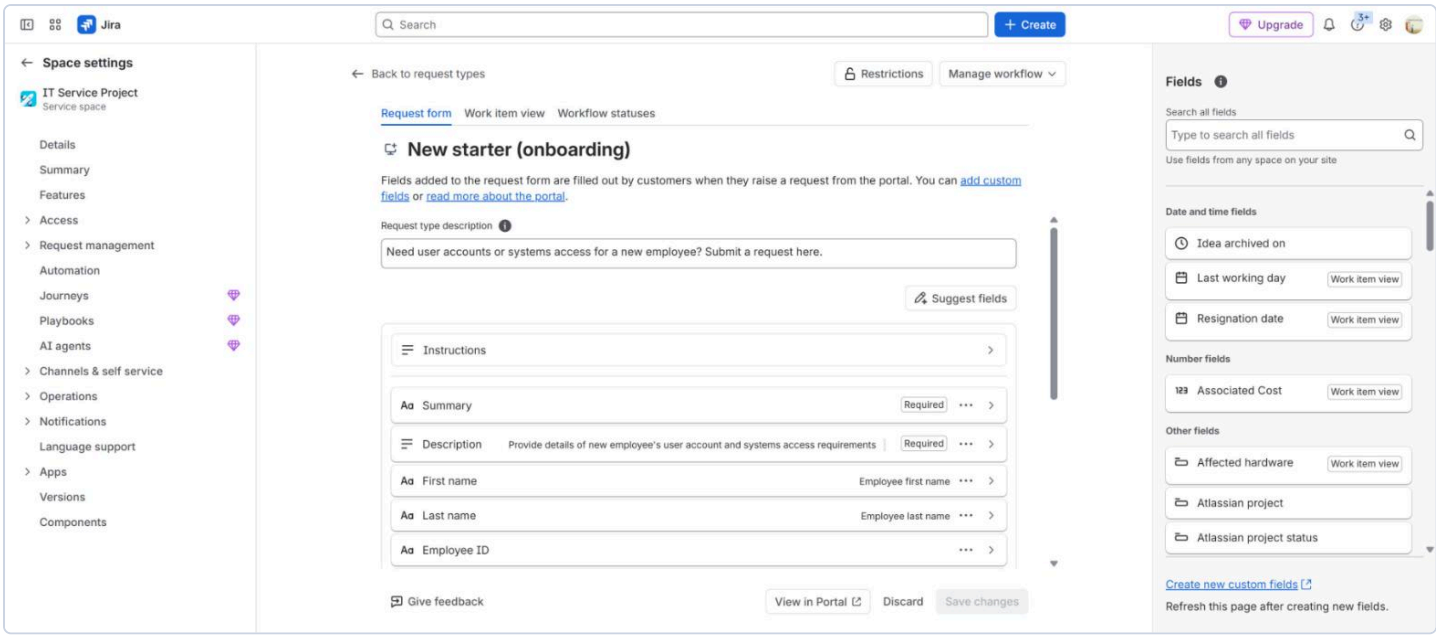
In the lab I created an object schema with one object type for laptops and one for people, and linked each laptop to the person holding it. The new starter request reads the laptop field from Assets, so the agent picks a real object instead of typing a model name by hand. When the ticket closes, the automation sets the status to **In use** and fills in the holder. The leaver request reverses it: status back to **Returned**, holder cleared.

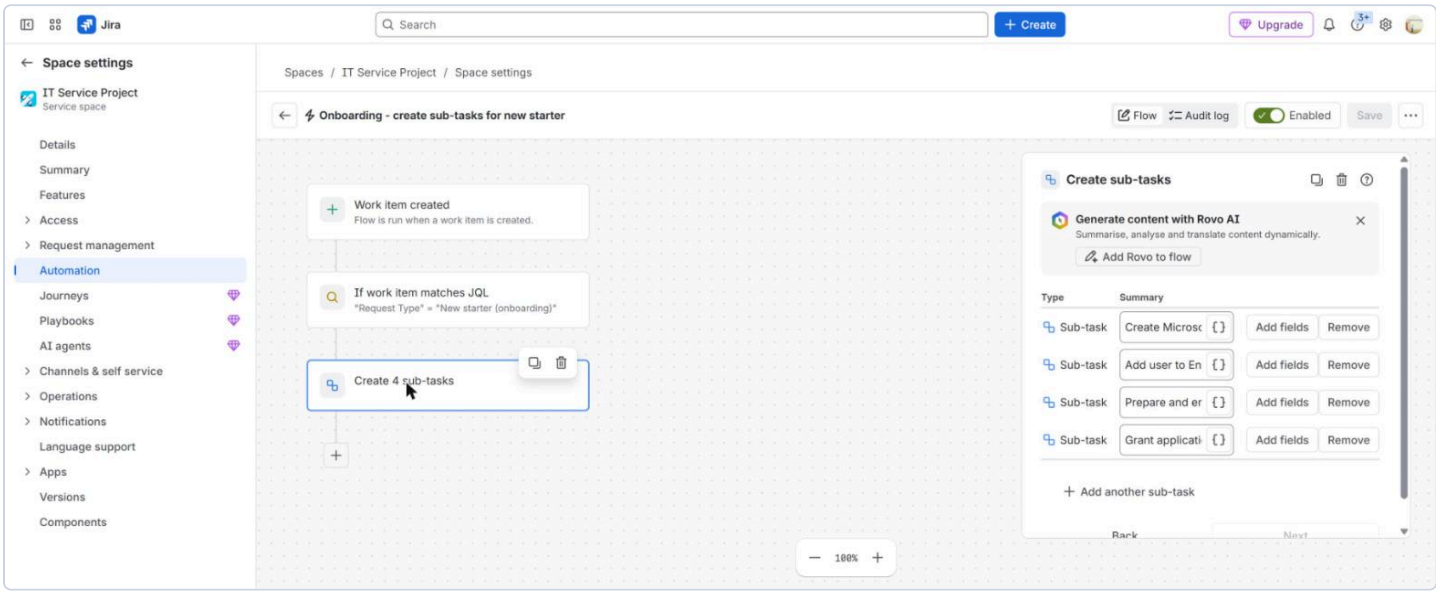
Example objects from the lab. These are invented for the lab and are not real people or real devices:

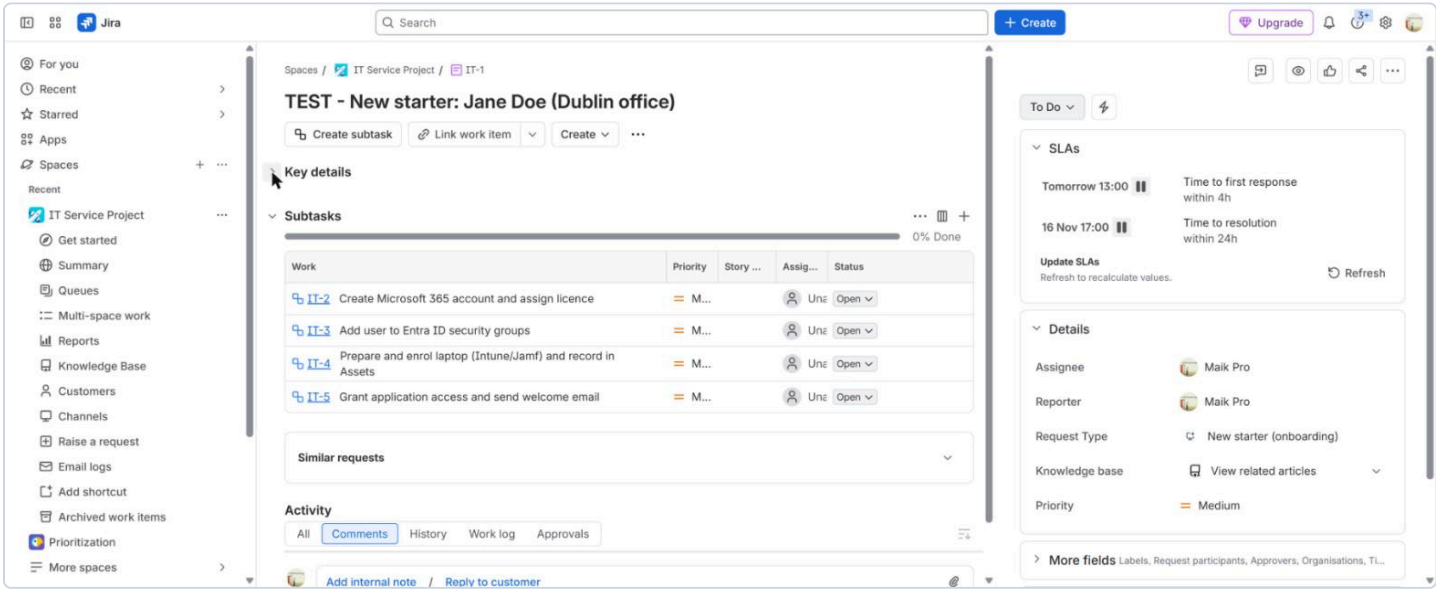
Asset tag	Model	Status	Assigned to
LAB-0001	MacBook Pro 14	In use	Aoife Byrne
LAB-0002	Dell Latitude 5440	In stock	unassigned
LAB-0003	MacBook Air 13	Returned	unassigned

This gives the same check I relied on at TikTok: the onboarding ticket cannot close while the asset record still says the laptop is in stock.

Proof







- Build log, Step 2: request forms
- Build log, Step 6: onboarding automation
- Joiner and leaver runbook
- Read-only access check: Get-UserAccessSnapshot.ps1
- Joiner automation with Microsoft Graph PowerShell

Tools

- Entra ID, Microsoft 365, Apple Business Manager, Puppet, BitLocker, Jira Service Management
- Imaging and build validation: MCU and a hardware checklist at TikTok
- Asset register: an internal ITAM system at TikTok, Jira Assets in my lab
- Familiar: same process, different system. Intune, Jamf and Mosyle for device enrolment. SSO and SCIM for app accounts.

Author: Marcus Paula
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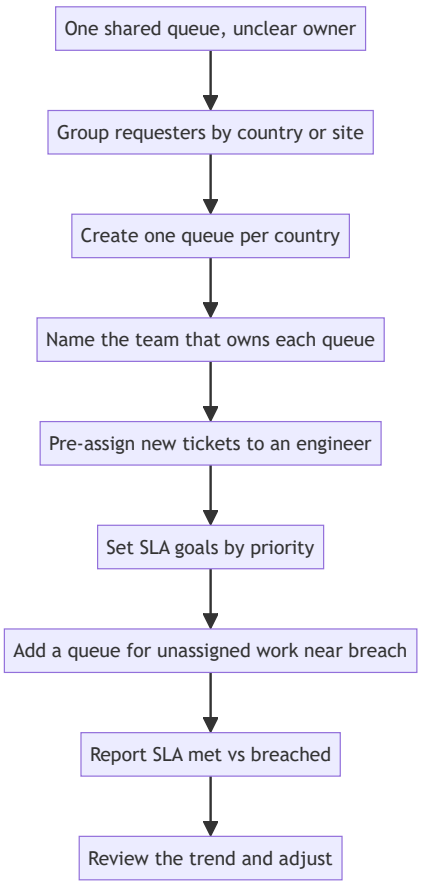
Service desk backlog and routing

The situation: Every ticket lands in one shared queue. Nobody is sure who owns what, and the backlog grows.

What I did in real work

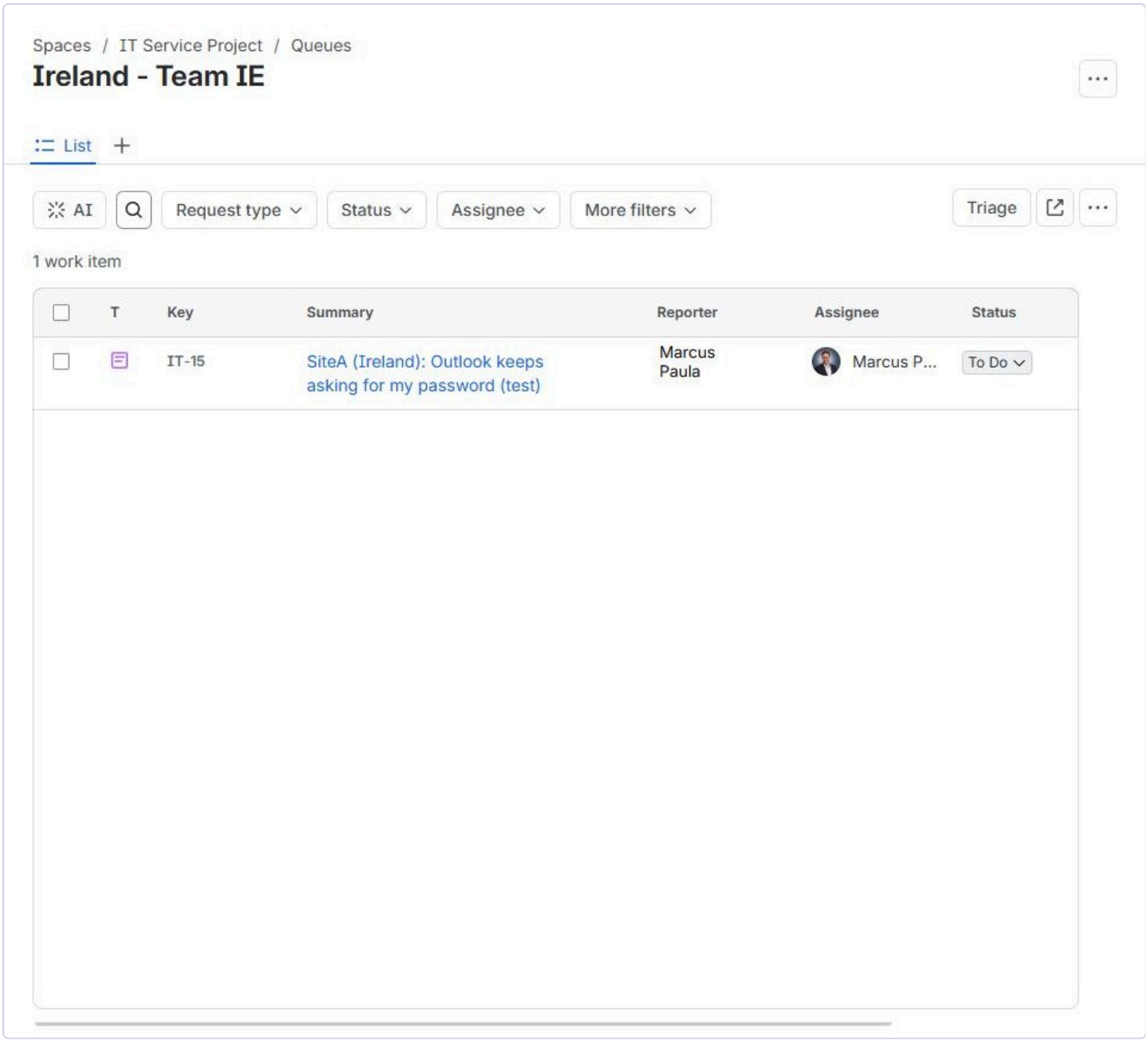
When I started at TikTok, all tickets went to one queue. Later the queue was split by country. Each country queue had a responsible team. Tickets were routed and pre-assigned to the technician, and SLAs were set by priority. After that, tickets were answered faster and ownership was clear.

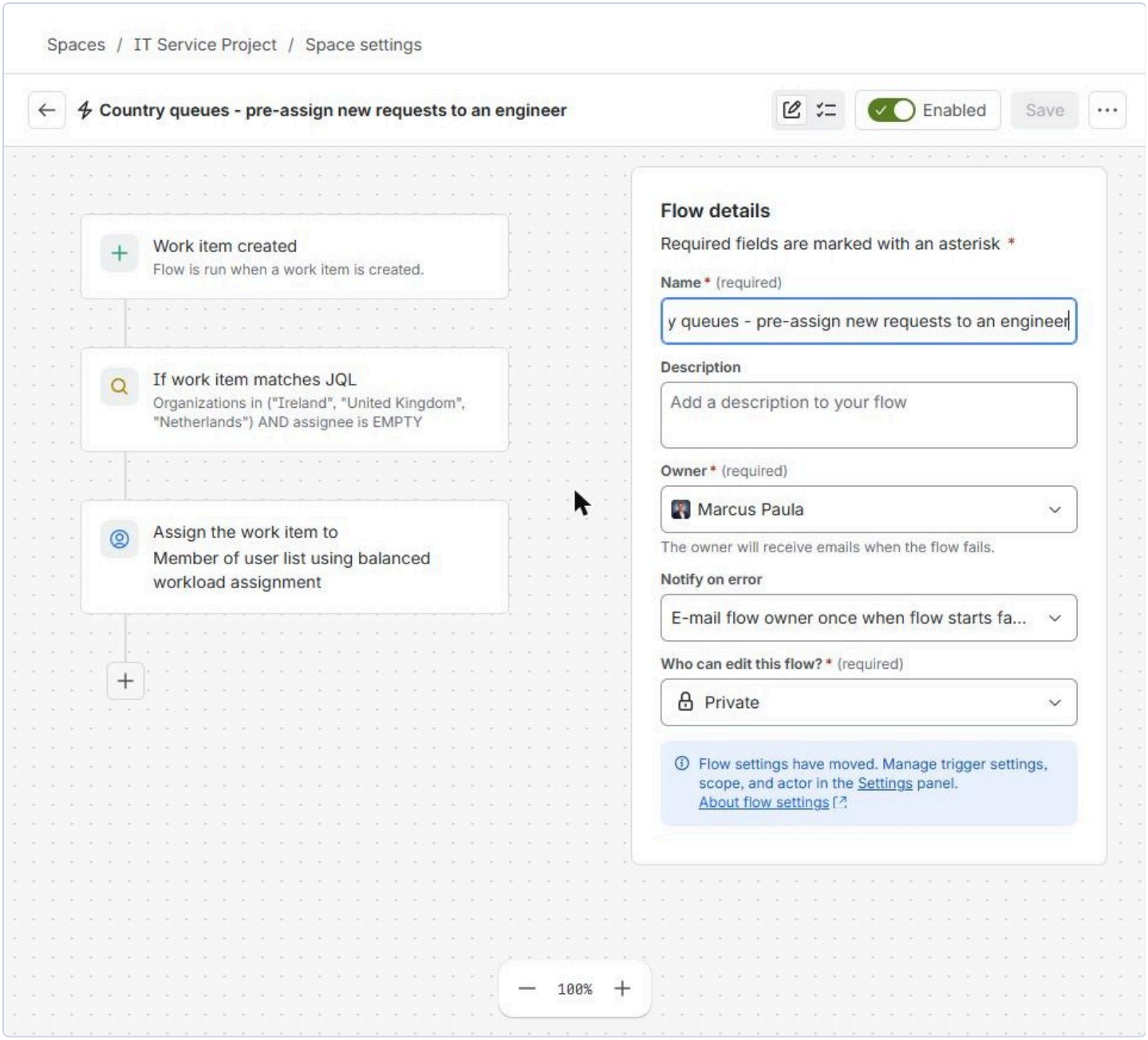
The process I follow

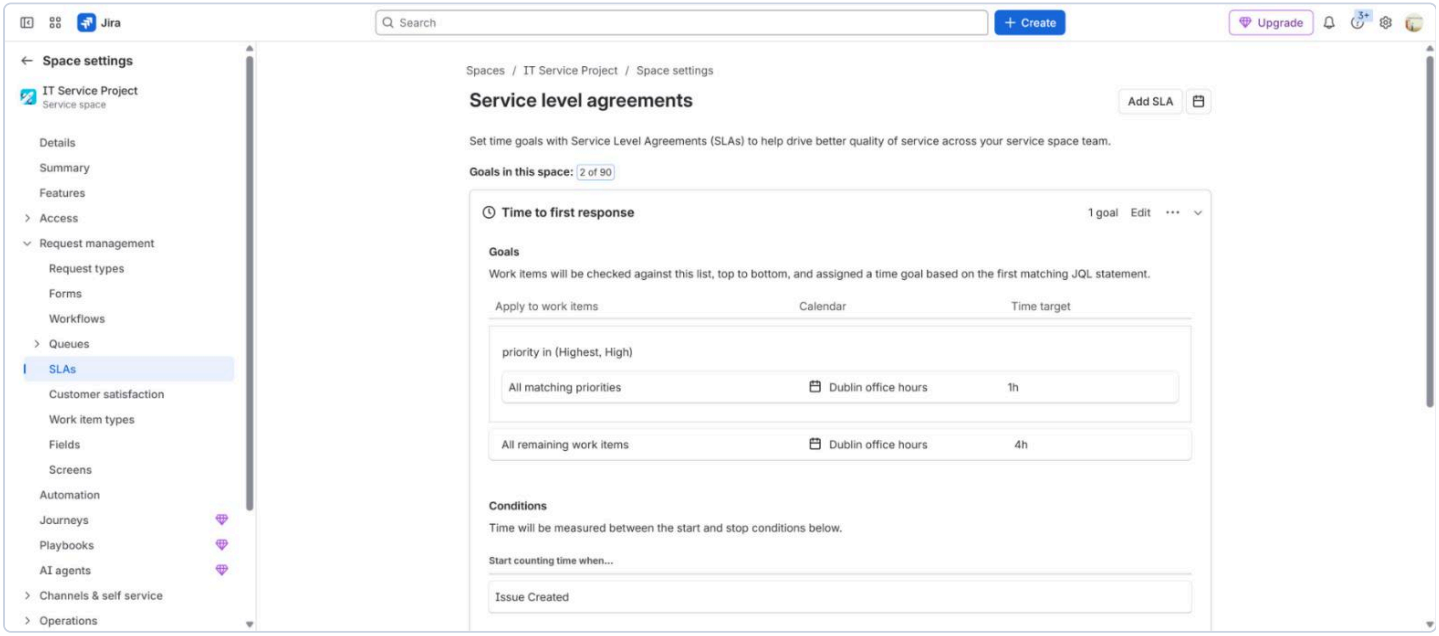


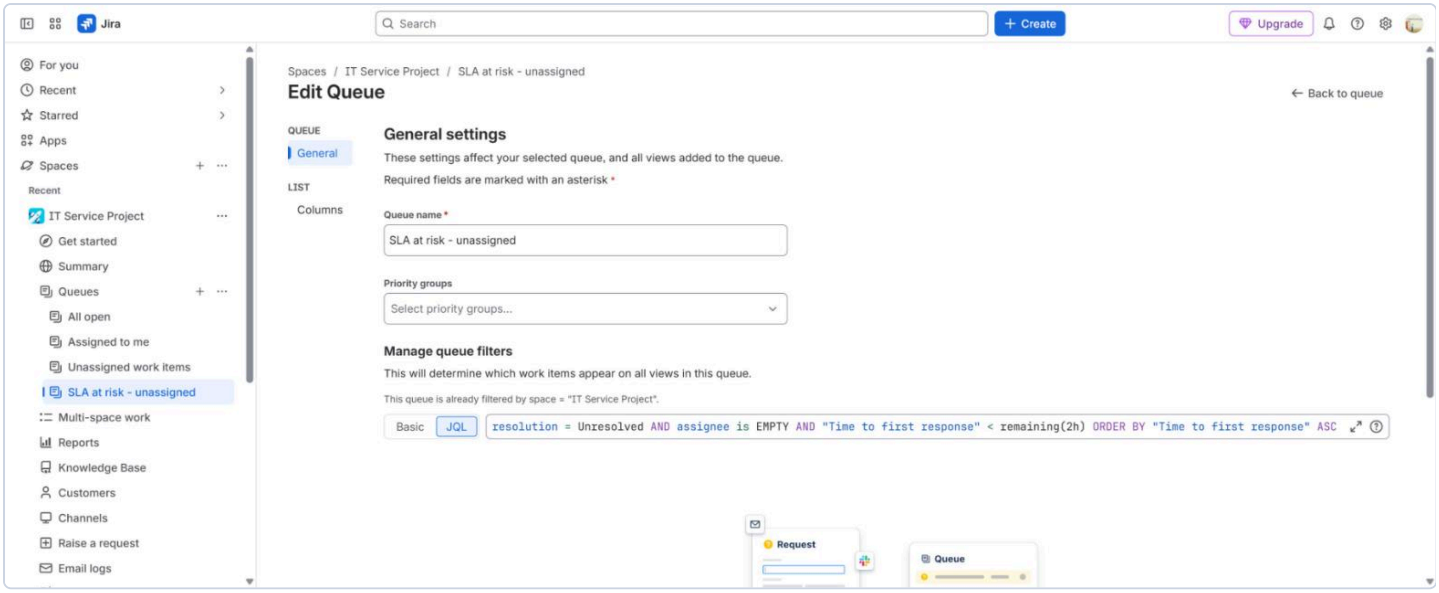
1. Look at the shared queue and list where the tickets come from.
2. Group requesters by country or site.
3. Create one queue per country.
4. Name the team that owns each queue.
5. Pre-assign each new ticket to an engineer. Balance the workload.
6. Set SLA goals by priority.
7. Add a queue that shows unassigned tickets close to breach.
8. Report SLA met vs breached every week.
9. Use the trend to change staffing or routing.

Proof


Spaces / IT Service Project / Queues
Ireland - Team IE
List
AI
Request type
Status
Assignee
More filters
Triage
1 work item
T
Key
Summary
Reporter
Assignee
Status
IT-15
SiteA (Ireland): Outlook keeps asking for my password (test)
Marcus Paula
Marcus P...
To Do


Spaces / IT Service Project / Space settings
Country queues - pre-assign new requests to an engineer
Enabled
Save
Work item created
Flow is run when a work item is created.
If work item matches JQL
Organizations in ("Ireland", "United Kingdom", "Netherlands") AND assignee is EMPTY
Assign the work item to
Member of user list using balanced workload assignment
Flow details
Required fields are marked with an asterisk *
Name * (required)
y queues - pre-assign new requests to an engineer
Description
Add a description to your flow
Owner * (required)
Marcus Paula
The owner will receive emails when the flow fails.
Notify on error
E-mail flow owner once when flow starts fa...
Who can edit this flow? * (required)
Private
Flow settings have moved. Manage trigger settings, scope, and actor in the Settings panel.
About flow settings
100%


Jira
Search
Create
Upgrade
Space settings
IT Service Project
Service space
Details
Summary
Features
Access
Request management
Request types
Forms
Workflows
Queues
SLAs
Customer satisfaction
Work item types
Fields
Screens
Automation
Journeys
Playbooks
AI agents
Channels & self service
Operations
Spaces / IT Service Project / Space settings
Service level agreements
Add SLA
Set time goals with Service Level Agreements (SLAs) to help drive better quality of service across your service space team.
Goals in this space: 2 of 90
Time to first response
1 goal
Edit
Goals
Work items will be checked against this list, top to bottom, and assigned a time goal based on the first matching JQL statement.
Apply to work items
Calendar
Time target
priority in (Highest, High)
All matching priorities
Dublin office hours
1h
All remaining work items
Dublin office hours
4h
Conditions
Time will be measured between the start and stop conditions below.
Start counting time when...
Issue Created



Spaces / IT Service Project / Reports /

SLA: first response met vs breached

Show

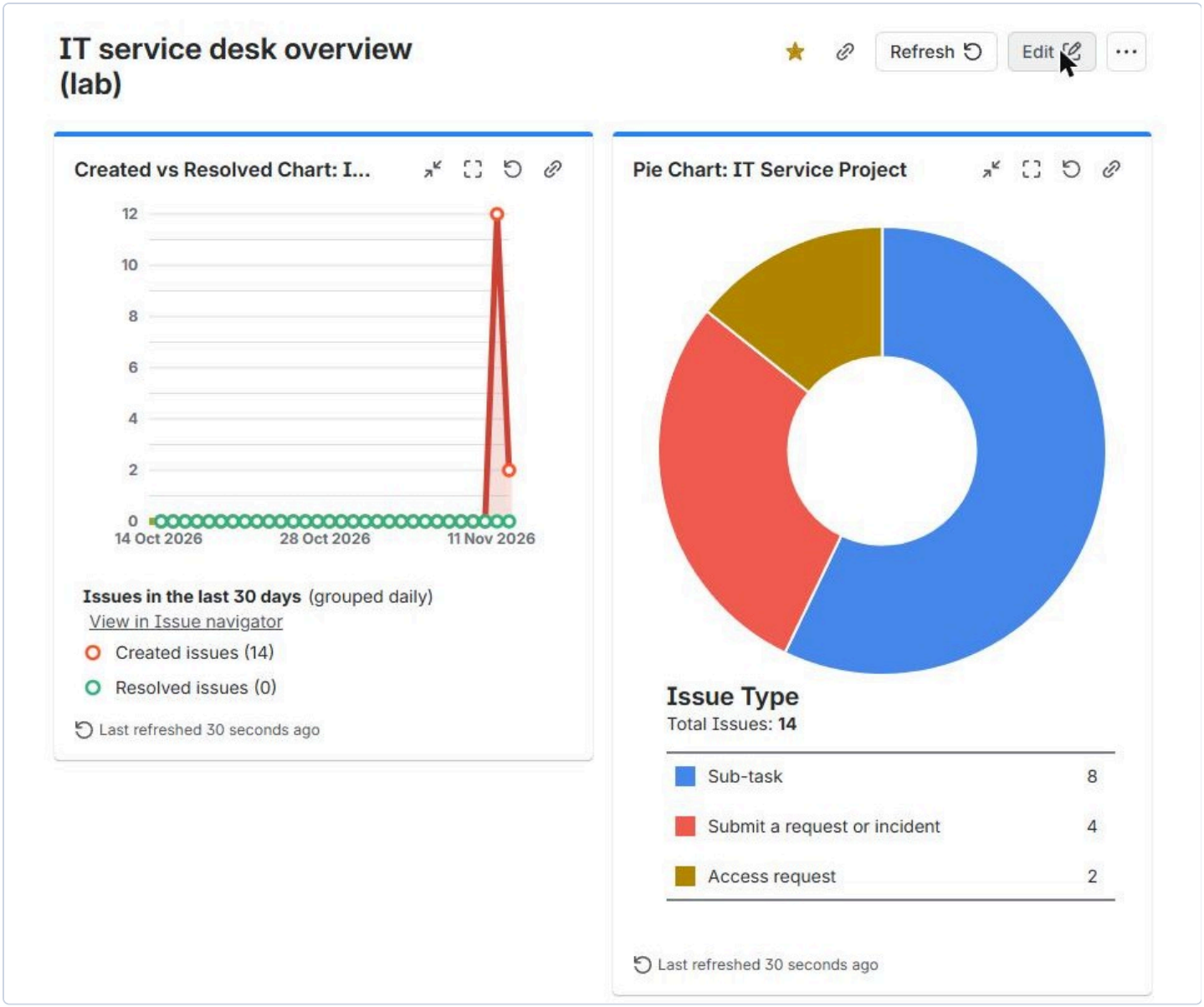
Past 7 days

0 Met

12 Breached



Date	Met	Breached
6 Nov 2026	0	0
7 Nov 2026	0	0
8 Nov 2026	0	0
9 Nov 2026	0	0



- Build log, Step 15: country queues and pre-assignment, tests IT-15, IT-16 and IT-17
- Build log, Step 4: SLAs
- Build log, Step 5: queue with JQL
- Build log, Step 14: report and dashboard
- Agent runbook: triage
- Phase 5: one service desk for three companies

Tools

- Jira Service Management: organizations, queues, JQL, automation, reports, dashboards

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Lab work uses synthetic data. No employer data is published.

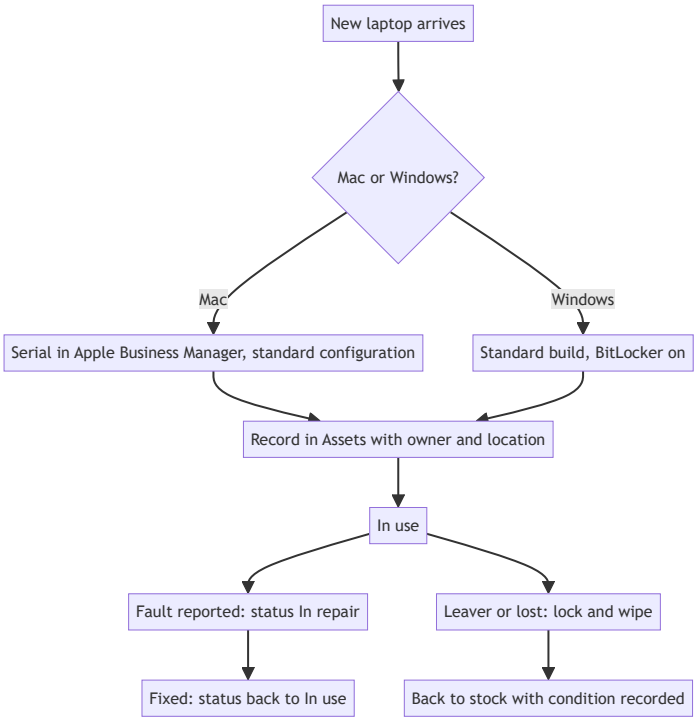
Mac and Windows device management

The situation: A company has Mac and Windows laptops. Each one must be enrolled, configured, encrypted and tracked, and wiped when it comes back.

What I did in real work

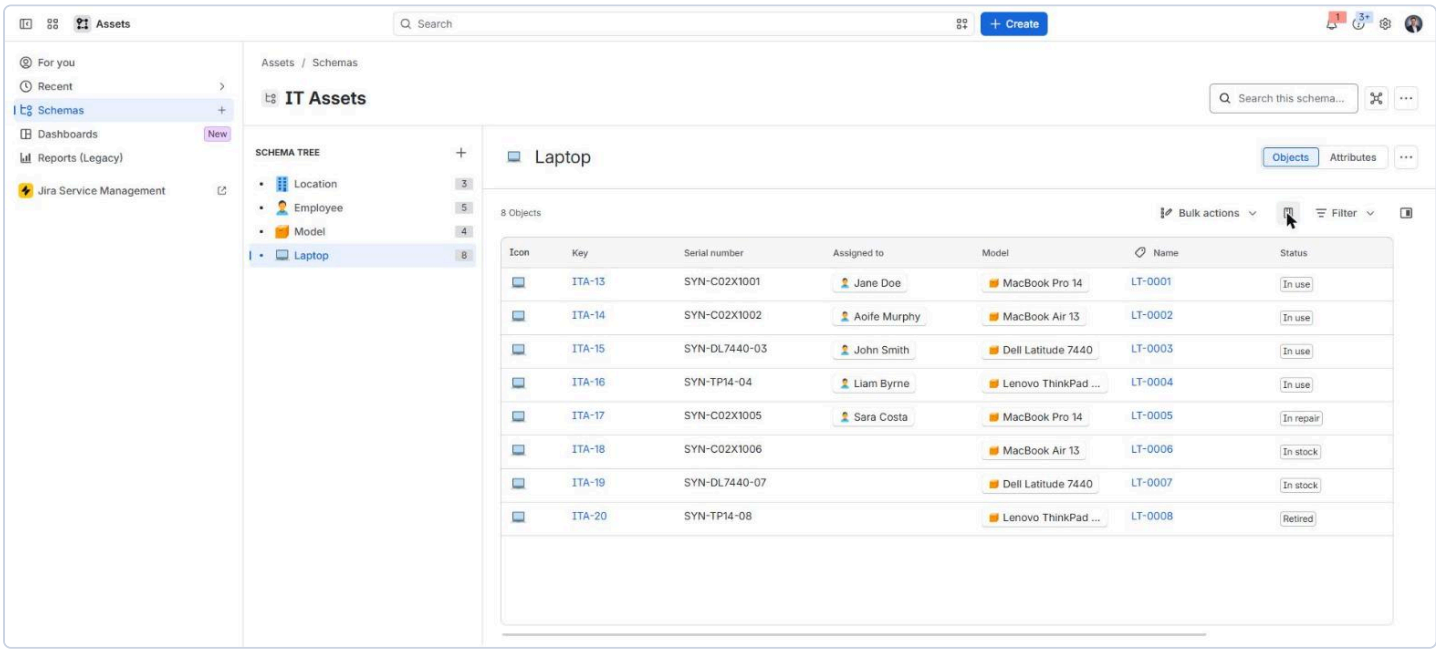
At TikTok I managed Macs with Apple Business Manager and Puppet, and Windows laptops with BitLocker and MBAM. I locked and wiped devices, and processed returns. I worked on the asset return process in three sites. I also cut imaging time from 60 to 15 minutes per device (scenario 15).

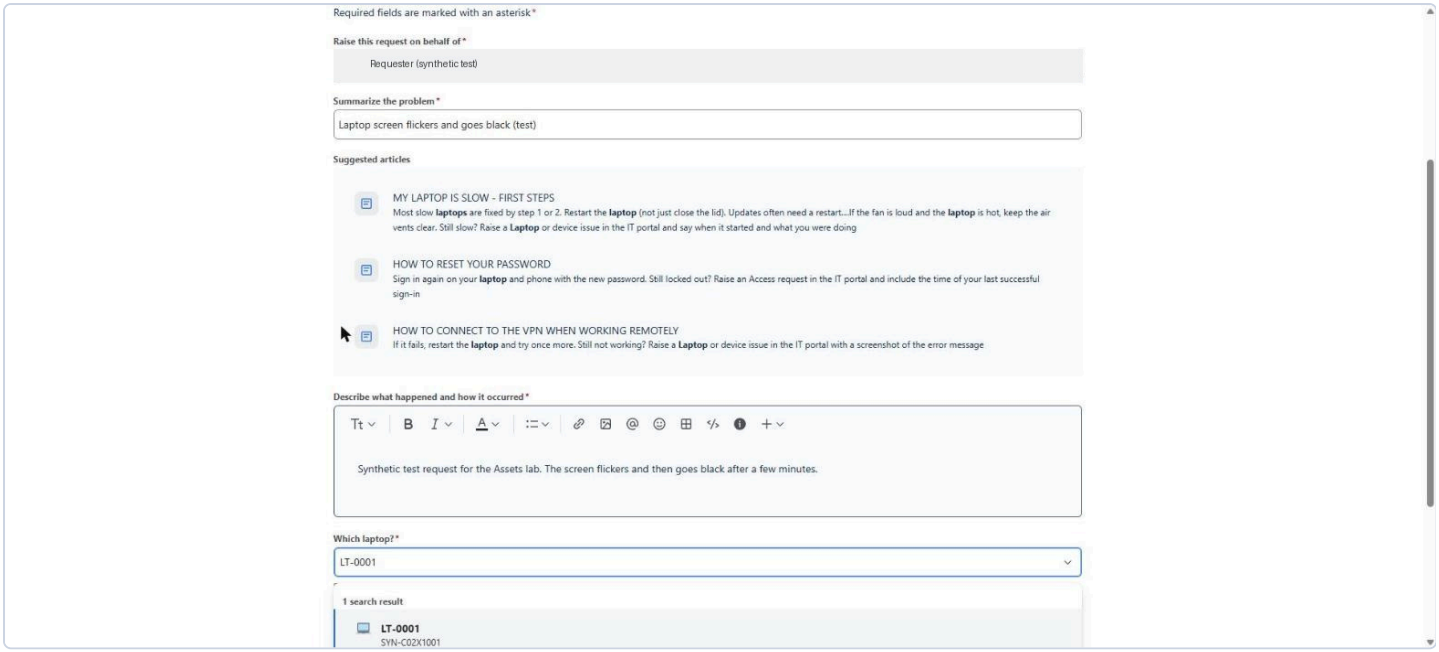
The process I follow

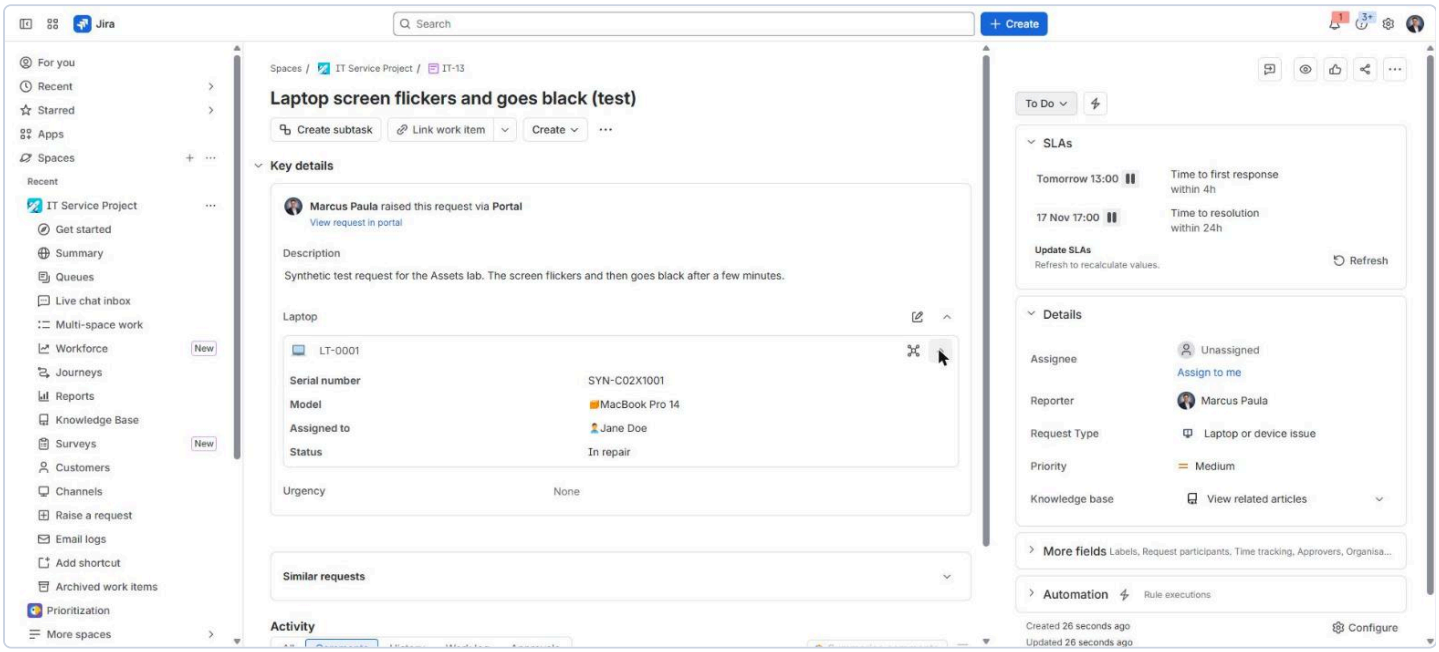


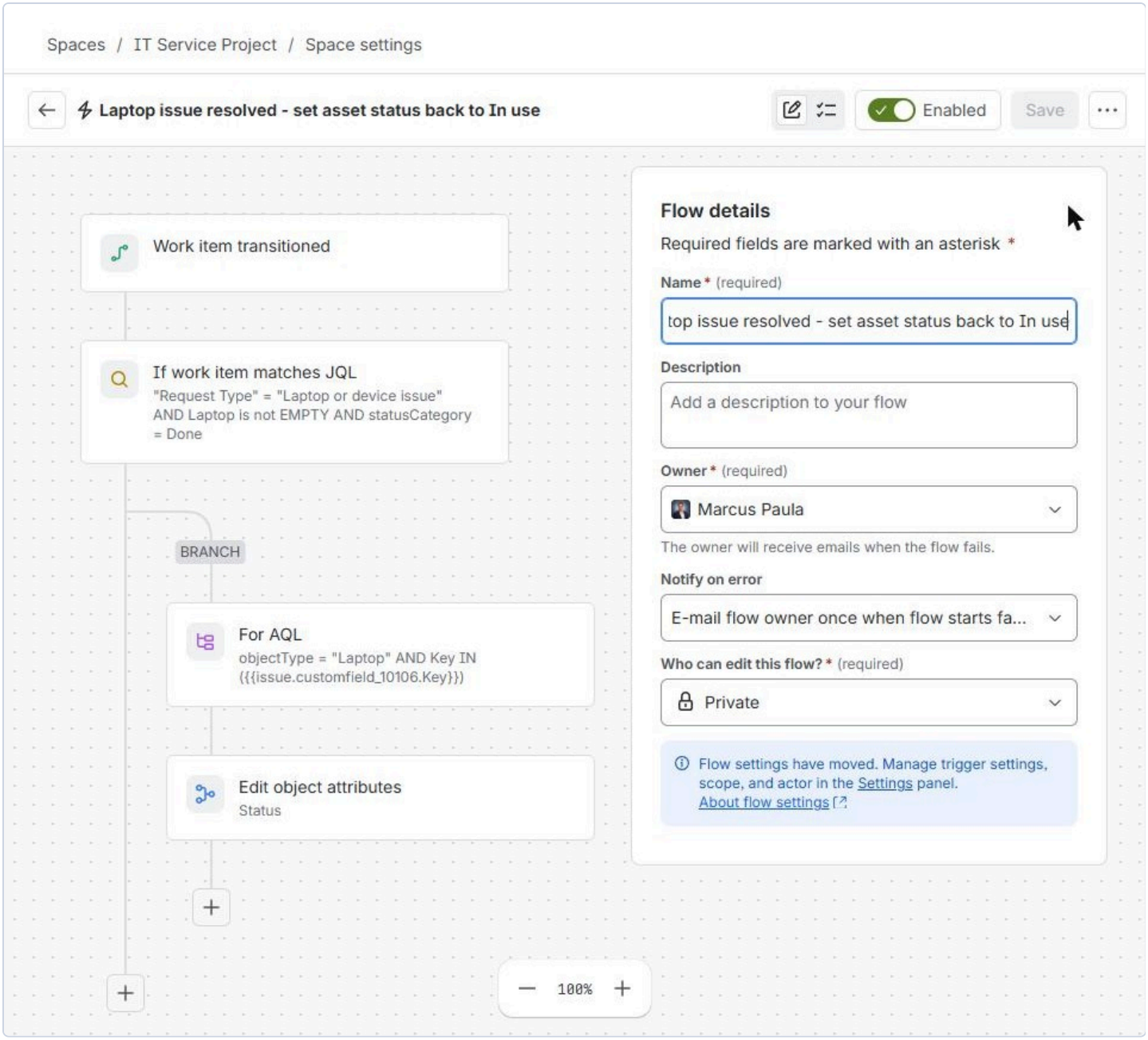
1. Receive the new laptop.
2. Mac: check the serial is in Apple Business Manager, then apply the standard configuration.
3. Windows: install the standard build and turn on BitLocker.
4. Record the laptop in Assets with owner, location, encryption and the tool that manages it.
5. Hand it over. The status is In use.
6. When a fault is reported, set the status to In repair.
7. When the fix is done, set the status back to In use.
8. For a leaver or a lost laptop, lock it first, then wipe it.
9. Put the returned laptop back in stock and record its condition.

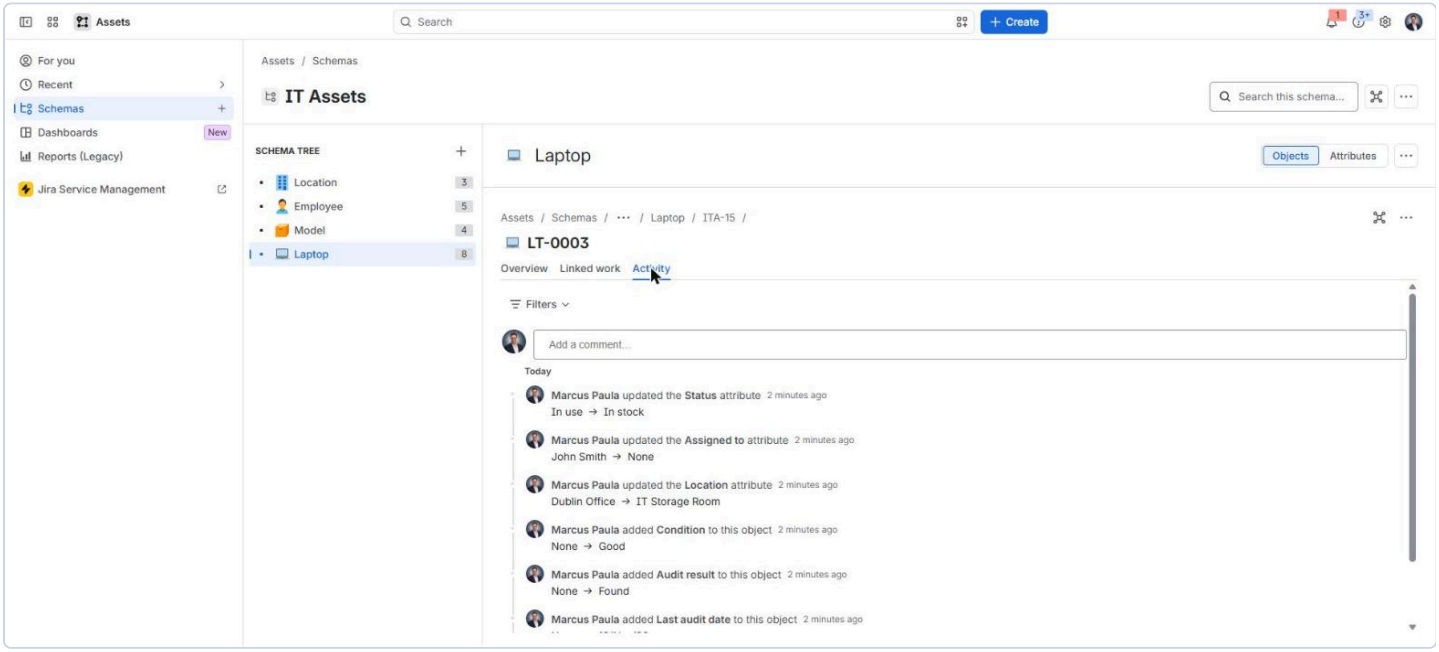
Proof




Required fields are marked with an asterisk*
Raise this request on behalf of *
Requester (synthetic test)
Summarize the problem *
Laptop screen flickers and goes black (test)
Suggested articles
MY LAPTOP IS SLOW - FIRST STEPS
Most slow laptops are fixed by step 1 or 2. Restart the laptop (not just close the lid). Updates often need a restart...If the fan is loud and the laptop is hot, keep the air vents clear. Still slow? Raise a Laptop or device issue in the IT portal and say when it started and what you were doing
HOW TO RESET YOUR PASSWORD
Sign in again on your laptop and phone with the new password. Still locked out? Raise an Access request in the IT portal and include the time of your last successful sign-in
HOW TO CONNECT TO THE VPN WHEN WORKING REMOTELY
If it fails, restart the laptop and try once more. Still not working? Raise a Laptop or device issue in the IT portal with a screenshot of the error message
Describe what happened and how it occurred *
Synthetic test request for the Assets lab. The screen flickers and then goes black after a few minutes.
Which laptop? *
LT-0001
1 search result
LT-0001
SYN-C02X1001


Jira
Search
Create
For you
Recent
Starred
Apps
Spaces
Recent
IT Service Project
Get started
Summary
Queues
Live chat inbox
Multi-space work
Workforce
New
Journeys
Reports
Knowledge Base
Surveys
New
Customers
Channels
Raise a request
Email logs
Add shortcut
Archived work items
Prioritization
More spaces
Spaces / IT Service Project / IT-13
Laptop screen flickers and goes black (test)
Create subtask
Link work item
Create
Key details
Marcus Paula raised this request via Portal
View request in portal
Description
Synthetic test request for the Assets lab. The screen flickers and then goes black after a few minutes.
Laptop
LT-0001
Serial number
SYN-C02X1001
Model
MacBook Pro 14
Assigned to
Jane Doe
Status
In repair
Urgency
None
Similar requests
Activity
To Do
SLAs
Tomorrow 13:00
Time to first response within 4h
17 Nov 17:00
Time to resolution within 24h
Update SLAs
Refresh to recalculate values.
Refresh
Details
Assignee
Unassigned
Assign to me
Reporter
Marcus Paula
Request Type
Laptop or device issue
Priority
Medium
Knowledge base
View related articles
More fields Labels, Request participants, Time tracking, Approvers, Organisa...
Automation
Rule executions
Created 26 seconds ago
Updated 26 seconds ago
Configure





- Project 08: tool mapping to Intune and Jamf
- Troubleshooting guide
- Build log, Step 11: Assets
- Build log, Step 15: laptop back to In use

- Project 04: asset return workflow
- Read-only endpoint check: Get-EndpointReadiness.ps1

Tools

- Apple Business Manager, Puppet, BitLocker, MBAM, Jira Service Management Assets
- Familiar: same process, different system. Intune, Jamf and Mosyle (enrol, configure, encrypt, lock, wipe).

Lab work uses synthetic data. No employer data is published.

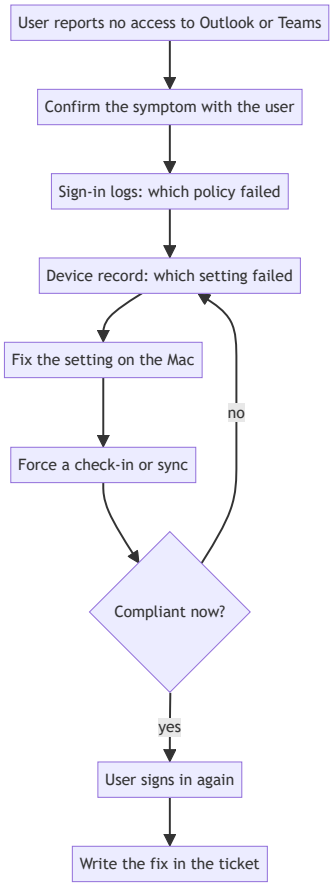
Non-compliant Mac loses Microsoft 365 access

The situation: A user on a Mac cannot open Outlook or Teams. The sign-in page says the device does not meet the company's requirements.

What I did in real work

At TikTok I supported Entra ID and Microsoft 365 sign-in problems. I worked with Macs managed by Apple Business Manager and Puppet, and Windows laptops with BitLocker and MBAM. Device compliance in Intune or Jamf was not my tool there. The process is the same: find the check that failed, fix the device, then sign in again.

The process I follow



1. Confirm the symptom with the user. Note the app, the time and the error text.
2. Open the Entra ID sign-in logs and find the policy that failed.
3. Open the device record and find the setting that failed. Common causes are encryption off, an old OS version or a device that has not checked in.
4. Fix that setting on the Mac.
5. Force a check-in or sync from the management tool.
6. Check the device is compliant now.
7. Ask the user to sign in again.
8. Write the cause and the fix in the ticket. If it repeats, write a knowledge base article.

Do not exclude the user from the policy to make the error go away. Fix the cause.

Proof

Process only, no screenshot in the lab for this tool. The closest proof:

- Troubleshooting guide: sign-in blocked by Conditional Access, device not compliant, Mac not checking in
- Read-only script that shows a user's managed devices and their compliance state: Get-UserAccessSnapshot.ps1
- Project 08: tool mapping to Intune and Jamf
- How the user reports it in the lab: laptop picker on the request form and IT-13 with the laptop set to In repair

Tools

- Entra ID sign-in logs, Microsoft 365, Apple Business Manager, Puppet
- Familiar: same process, different system. Intune and Jamf compliance, Mosyle.

Lab work uses synthetic data. No employer data is published.

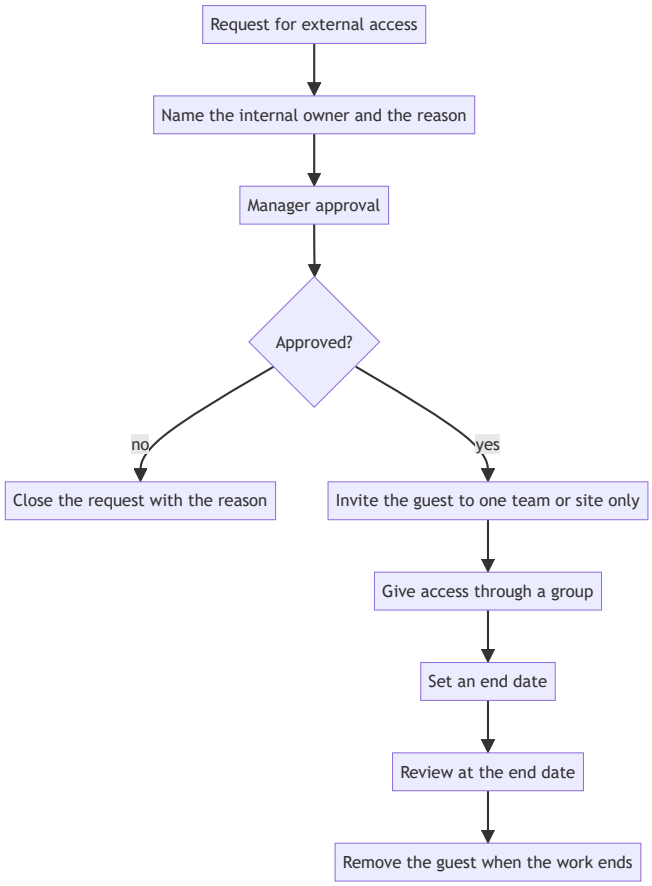
External collaboration in Microsoft 365

The situation: A team wants to share files and a Teams channel with an outside partner. The company must control what is shared and for how long.

What I did in real work

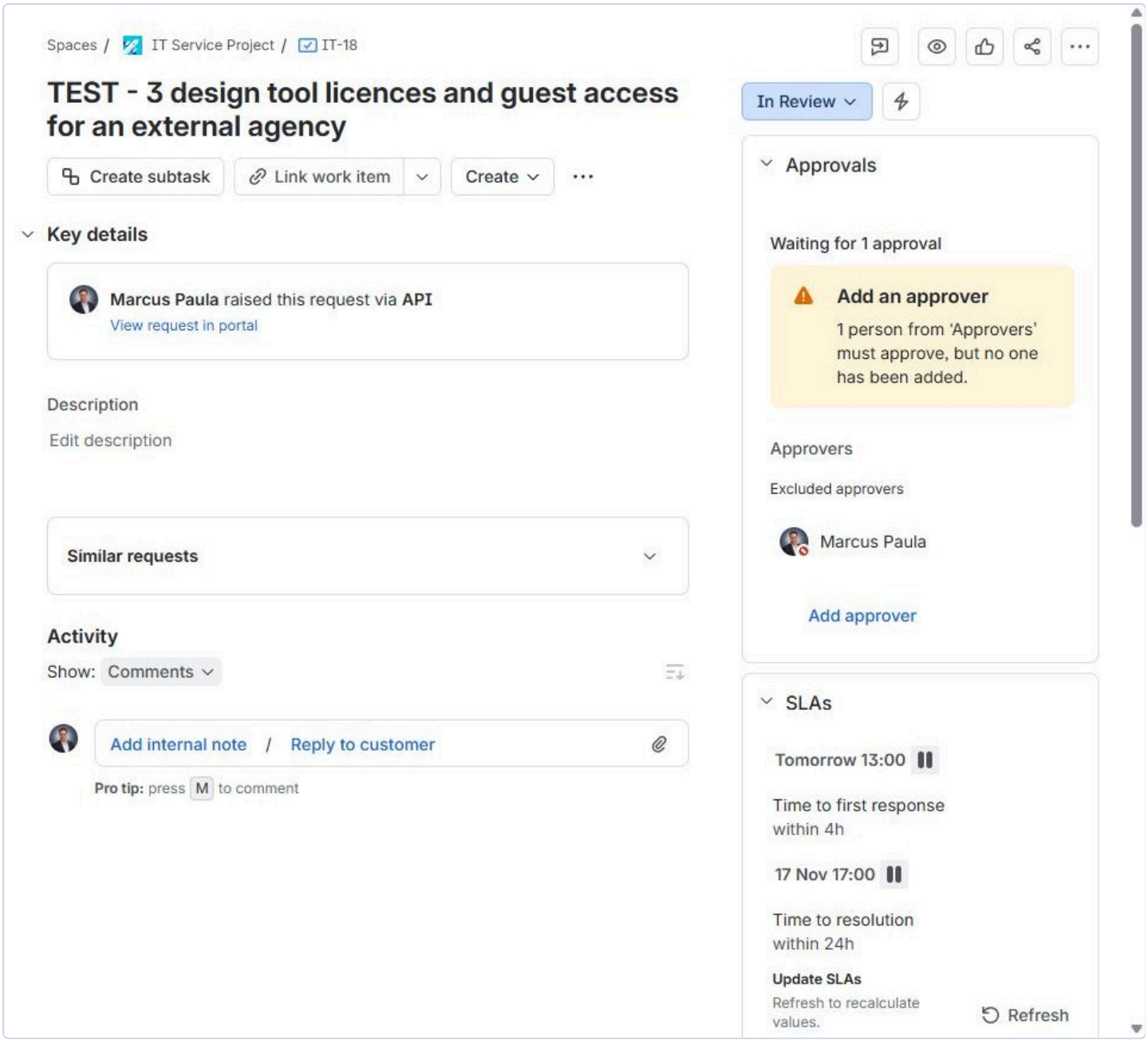
At TikTok I did not own the tenant sharing settings. I worked this with the owning team. My part was the access around it: Microsoft 365 licences, security groups, distribution lists and shared mailboxes, with access given through groups.

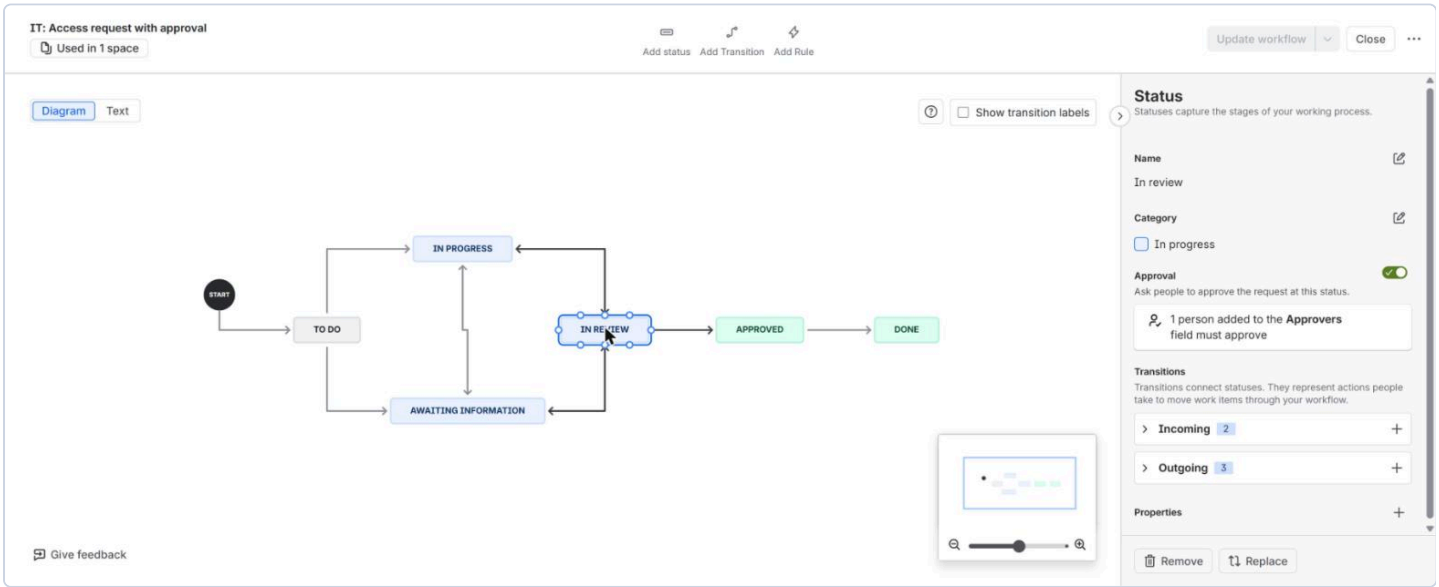
The process I follow

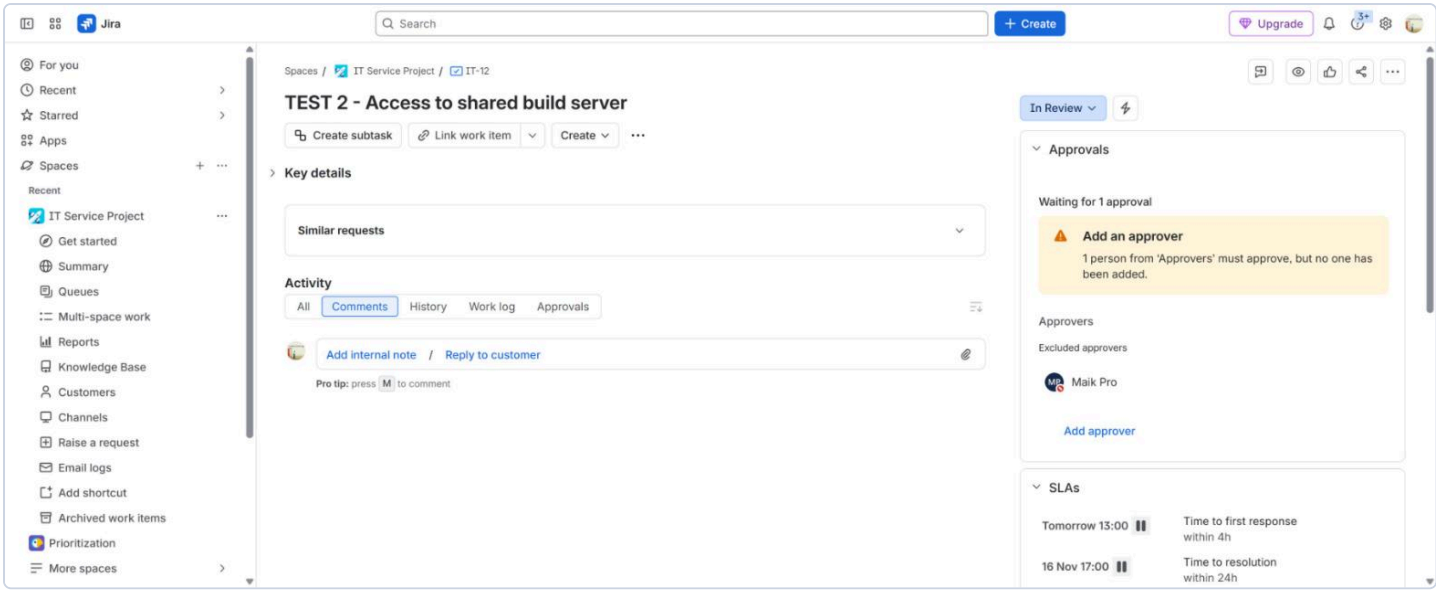


1. The employee raises a request for external access.
2. Record the internal owner, the partner and the business reason.
3. Send the request for manager approval. The requester cannot approve it.
4. If it is rejected, close the request and record the reason.
5. Invite the guest to one team or one site only, not to the whole tenant.
6. Give access through a group, so it is easy to review and remove.
7. Set an end date for the access.
8. At the end date, ask the owner if the access is still needed.
9. Remove the guest when the work ends.

Proof


Spaces / IT Service Project / IT-18
TEST - 3 design tool licences and guest access for an external agency
In Review
Create subtask
Link work item
Create
Key details
Marcus Paula raised this request via API
View request in portal
Description
Edit description
Similar requests
Activity
Show: Comments
Add internal note / Reply to customer
Pro tip: press M to comment
Approvals
Waiting for 1 approval
Add an approver
1 person from 'Approvers' must approve, but no one has been added.
Approvers
Excluded approvers
Marcus Paula
Add approver
SLAs
Tomorrow 13:00
Time to first response within 4h
17 Nov 17:00
Time to resolution within 24h
Update SLAs
Refresh to recalculate values.
Refresh


IT: Access request with approval
Used in 1 space
Add status
Add Transition
Add Rule
Update workflow
Close
Diagram
Text
Show transition labels
START
TO DO
IN PROGRESS
IN REVIEW
AWAITING INFORMATION
APPROVED
DONE
Status
Statuses capture the stages of your working process.
Name
In review
Category
In progress
Approval
Ask people to approve the request at this status.
1 person added to the Approvers field must approve
Transitions
Transitions connect statuses. They represent actions people take to move work items through your workflow.
Incoming 2
Outgoing 3
Properties
Remove
Replace
Give feedback



- Build log, Step 15: SaaS licence or external guest access request
- Build log, Step 8: workflow and approvals
- Project 08: licences, groups and mailboxes

The Microsoft 365 sharing settings are not in the lab.

Tools

- Entra ID guest users, Microsoft 365 groups, Teams, SharePoint, Jira Service Management approvals
- Familiar: same process, different system. At TikTok we used Lark. Guest access in similar apps such as Teams, Slack and Miro follows the same process.

Lab work uses synthetic data. No employer data is published.

Joiner, mover and leaver automation

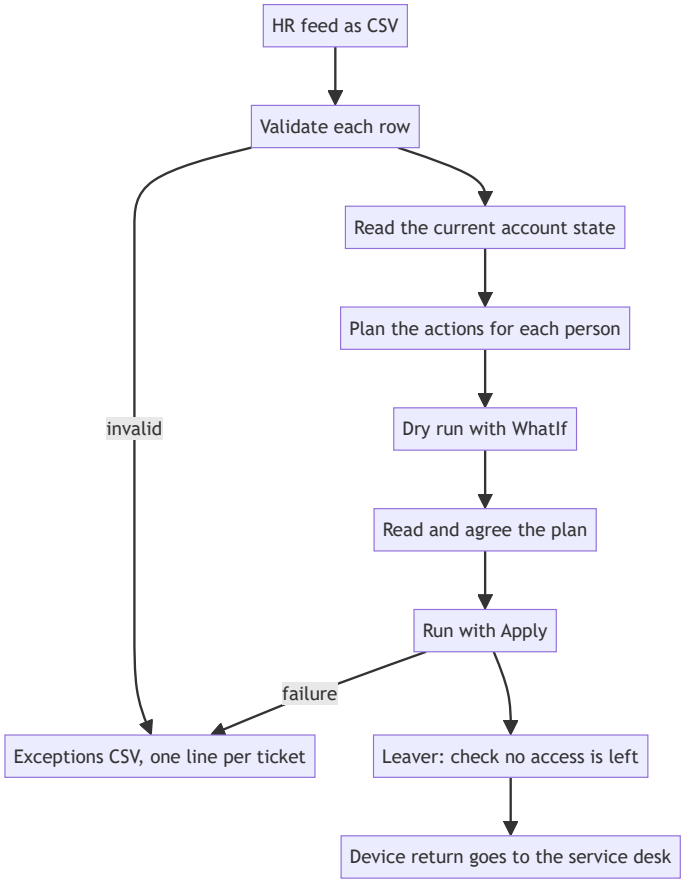
The situation: IT creates and removes accounts by hand from HR emails. Steps get missed, and some leavers keep access.

What I did in real work

At TikTok I ran joiner and leaver steps in Entra ID and Microsoft 365, and the device steps with Apple Business Manager, Puppet and BitLocker / MBAM. I created and disabled accounts, reset passwords and MFA, assigned licences, and managed security groups, distribution lists and shared mailboxes. For devices I did the lock, wipe and return with Apple Business Manager and Puppet for Macs, and BitLocker and MBAM for Windows.

In a lab I automated the same steps with Microsoft Graph PowerShell. I did not build an HR integration at TikTok.

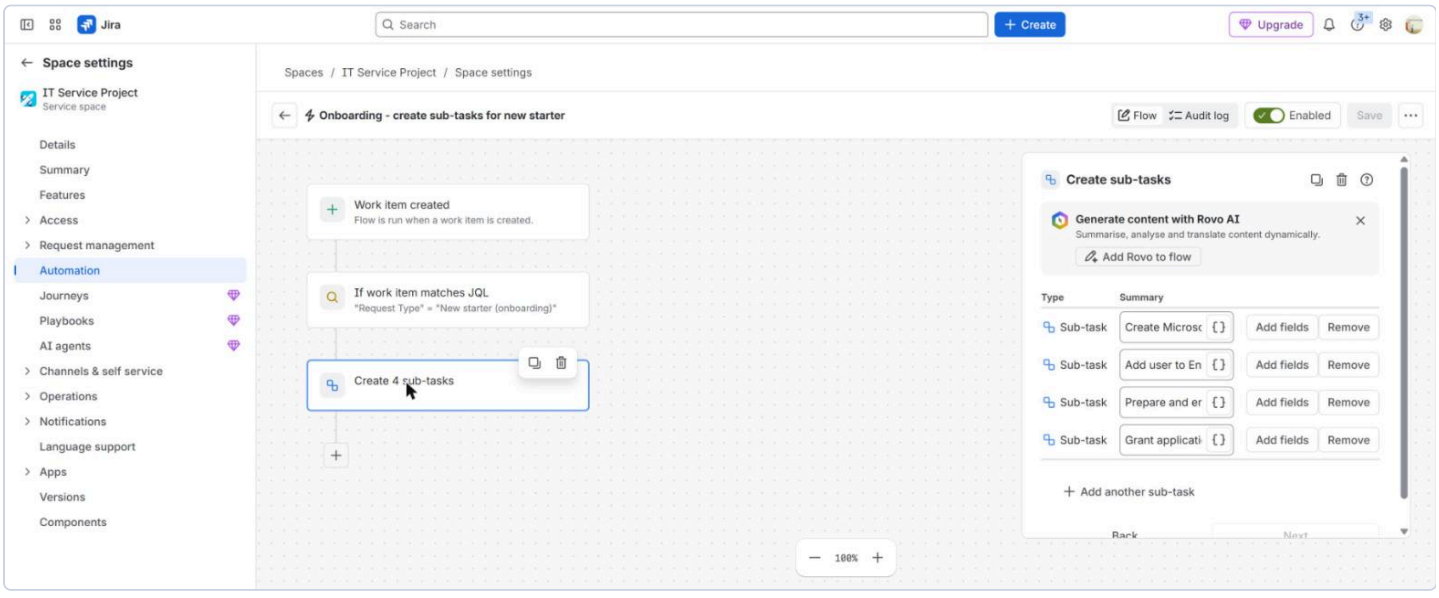
The process I follow

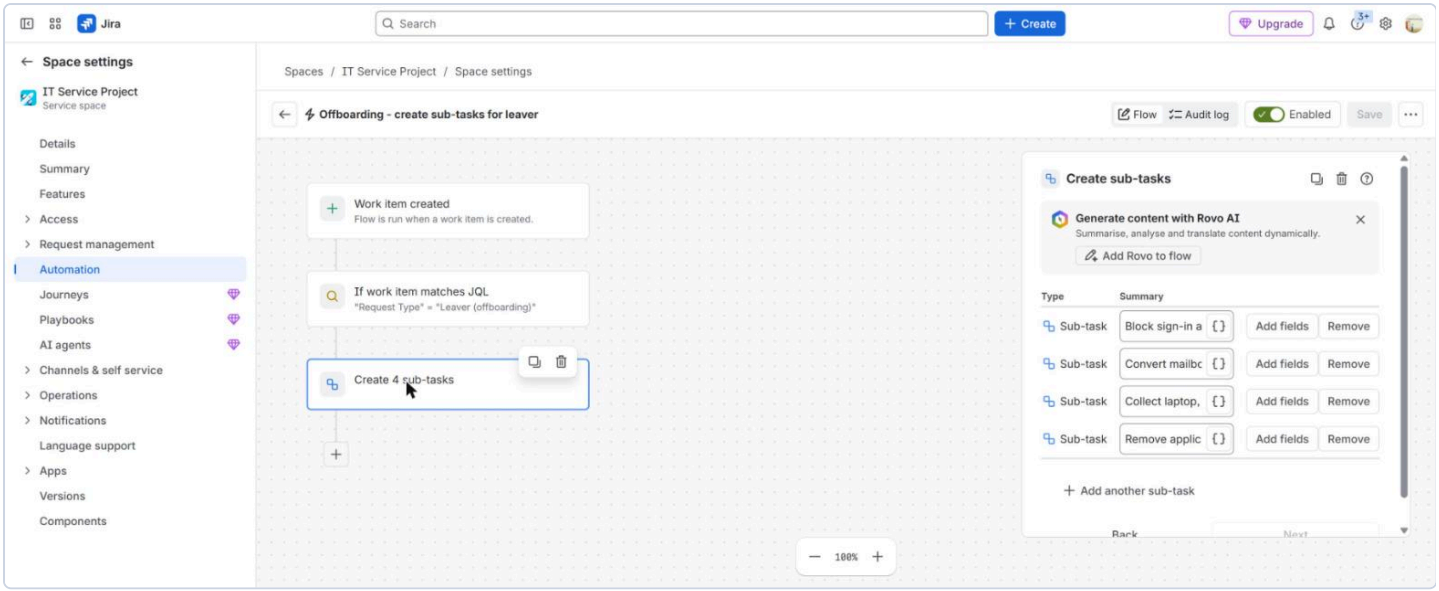


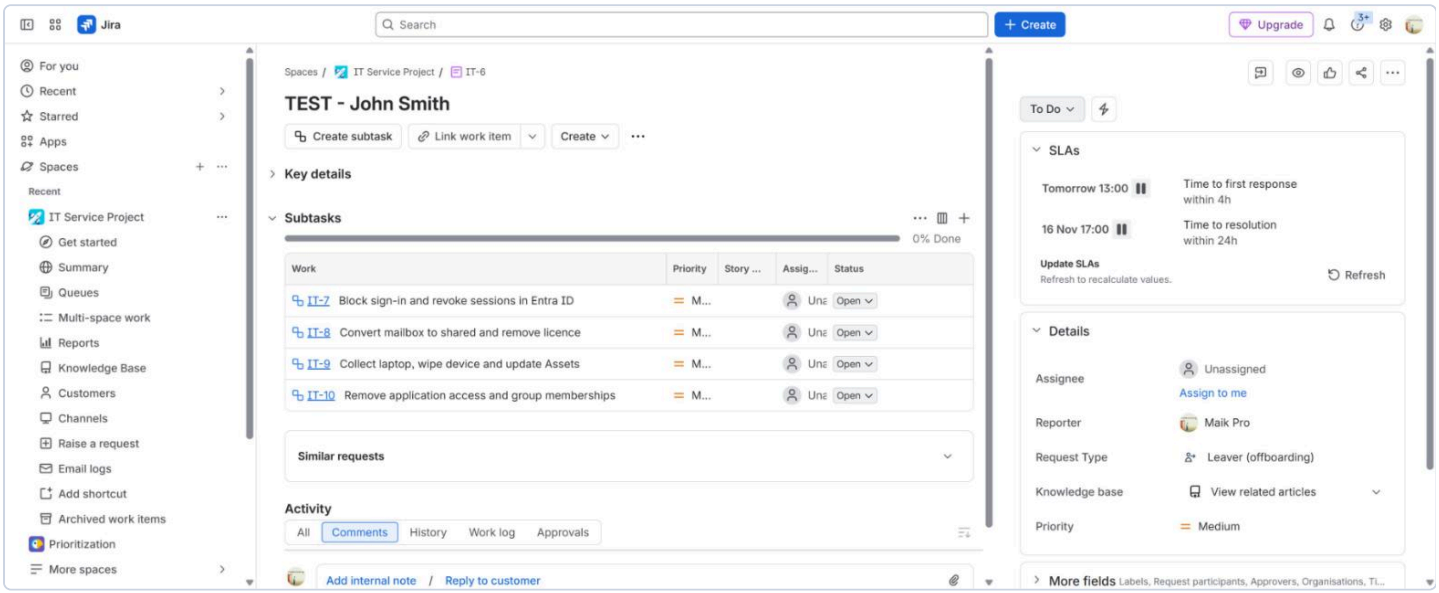
1. Take the HR feed as a CSV file.
2. Validate each row. Send bad rows to the exceptions file.
3. Read the current state of each account.
4. Plan the actions. Joiner: create, usage location, manager, groups, licence. Mover: add the new group, remove the old one. Leaver: block sign-in and revoke sessions first.
5. Run a dry run with `-WhatIf`. Nothing changes.
6. Read the plan and agree it.
7. Run with `-Apply`.
8. Turn every failure into one line in the exceptions file. A leaver failure is High priority.
9. For each leaver, run the access check.
10. Send the device return list to the service desk.

Urgent leavers do not wait for the feed. The service desk blocks sign-in by hand.

Proof







- Project 09: Graph PowerShell module and how to run it
- Project 09: module source
- Project 09: 31 Pester tests with mocked Graph calls
- CI workflow that runs the tests
- Synthetic HR feed
- Joiner and leaver runbook
- Build log, Step 7: offboarding automation
- Active Directory automation repository: onboard, transfer and offboard scripts for on-premises Active Directory, with Pester tests

Tools

- Microsoft Graph PowerShell, Exchange Online PowerShell, Pester, GitHub Actions, Active Directory PowerShell
- Familiar: same process, different system. SSO and SCIM provisioning from Entra ID to SaaS apps.

Lab work uses synthetic data. No employer data is published.

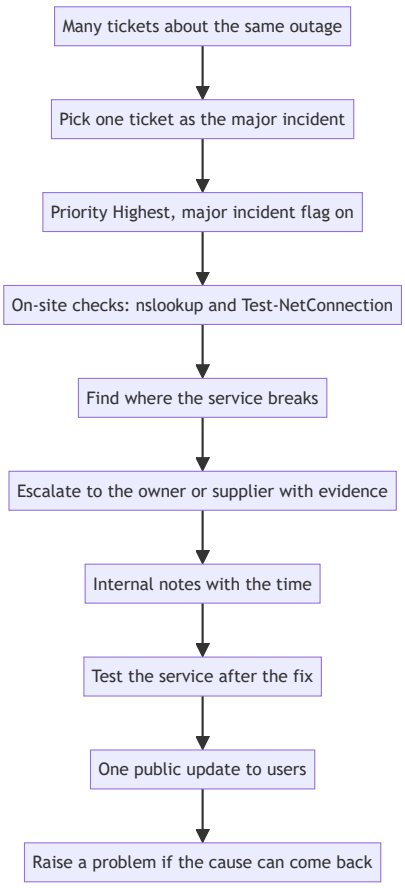
Office internet outage: the first 60 minutes

The situation: At 09:00 the office loses internet. Nobody can connect to the VPN, and tickets start to arrive from many people.

What I did in real work

At TikTok the firewall and VPN were owned by the network team. My part was on site. I checked where the service broke with `nslookup` and `Test-NetConnection`, and I escalated to the network team with that evidence.

The process I follow



Minute 0 to 10: declare it

1. Pick one ticket as the major incident. Keep the other tickets open and link them.
2. Set priority Highest and turn on the major incident flag.

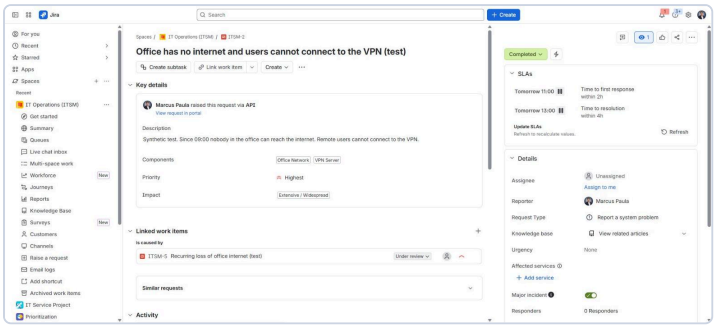
Minute 10 to 30: find the fault and escalate

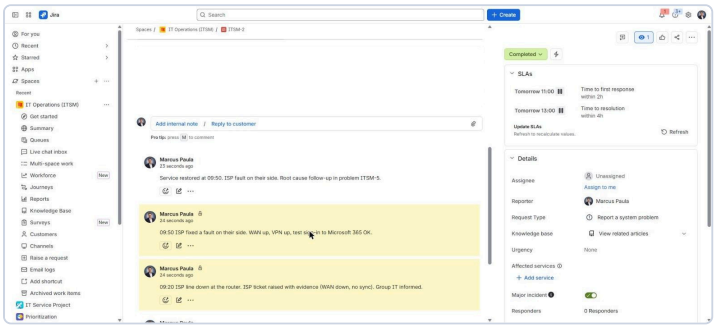
1. Check name resolution: `nslookup example.com`
2. Check the connection to a service and port: `Test-NetConnection example.com -Port 443`
3. Check the router or line status on site.
4. Raise a ticket with the owning team or the supplier. Attach the evidence.
5. Write each finding as an internal note with the time.

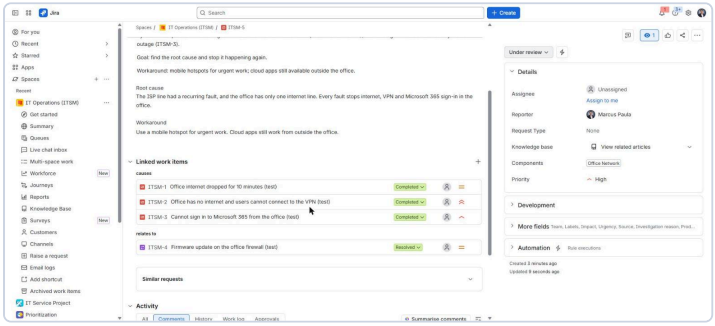
Minute 30 to 60: restore and confirm

1. When the owner reports a fix, test it yourself: internet, VPN and a Microsoft 365 sign-in.
2. Send one public update: what is back, when, the cause in one line and the problem reference.
3. Resolve the major incident and the linked tickets.
4. Raise a problem if the cause can come back.

Proof







- Agent runbook: major incident, the first 60 minutes
- Agent runbook: hand-off to problem management
- Build log, Step 12: incident, problem and change
- Knowledge base article on VPN, linked to the portal

Tools

- Jira Service Management incidents and problems, `nslookup`, `Test-NetConnection`
- Familiar: same process, different system. FortiGate firewall and VPN.

Lab work uses synthetic data. No employer data is published.

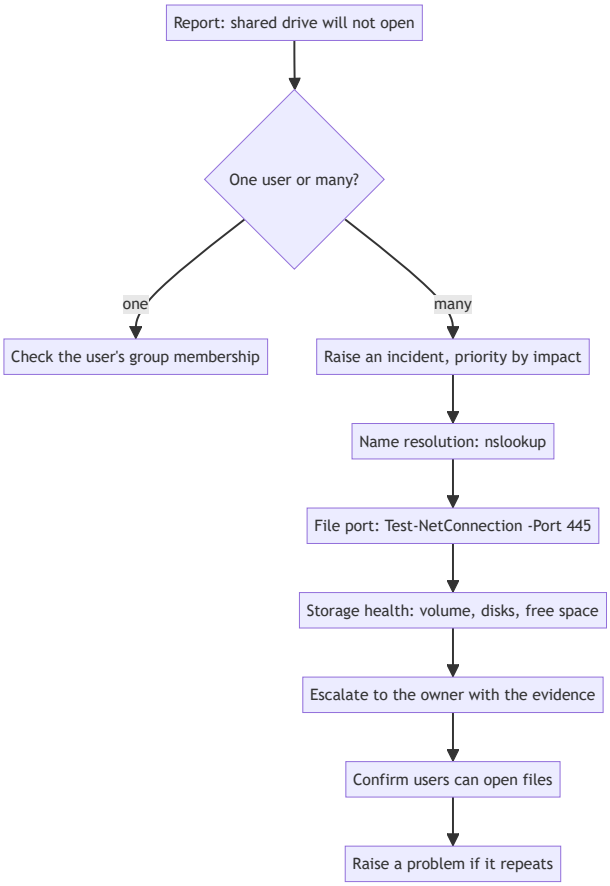
Shared file service recovery

The situation: Staff in the office cannot open the shared drive. The cause could be storage, network or permissions.

What I did in real work

At TikTok I worked this with the owning team. My part was on site: I checked the network path with `nslookup` and `Test-NetConnection`, and I escalated with that evidence to the team that owned the service.

The process I follow



1. Ask the user which share, which error and since when.
2. Check if one user or many users are affected.
3. One user: check the user's group membership for that share.
4. Many users: raise an incident and set the priority by impact.
5. Check name resolution: `nslookup fileserver.example.com`
6. Check the file sharing port: `Test-NetConnection fileserver.example.com -Port 445`
7. Ask the owner to check storage health: volume status, disk status and free space.
8. Escalate to the owning team with the results of steps 5 to 7.
9. After the fix, confirm users can open and save files.
10. If the same fault comes back, raise a problem and link the incidents.

Proof

Process only, no screenshot in the lab for this tool. The closest proof:

- The same incident and problem pattern, from the lab outage: major incident ITSM-2, timeline and problem ITSM-5
- Agent runbook: major incident and hand-off to problem
- Read-only script that lists a user's groups: Get-UserAccessSnapshot.ps1
- Generic storage and network validation checklist

Tools

- `nslookup`, `Test-NetConnection`, Entra ID groups, Jira Service Management incidents and problems
- Familiar: same process, different system. QNAP storage.

Lab work uses synthetic data. No employer data is published.

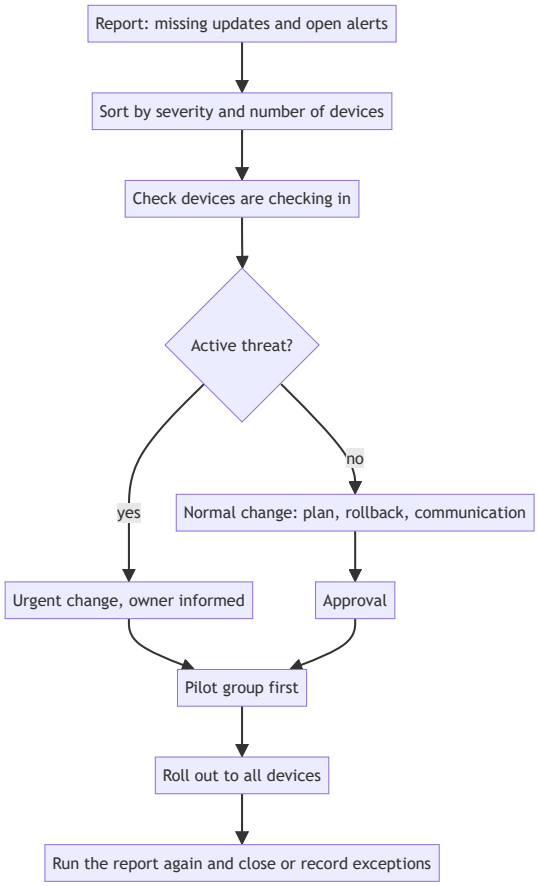
Missing patches and open security alerts

The situation: A report shows many laptops missing updates and a list of open security alerts. Nobody has a plan to close them.

What I did in real work

At TikTok I kept device images current. I tested the image every day, and the standard checklist after imaging included Windows updates and vendor updates. For security alerts, I worked this with the owning team.

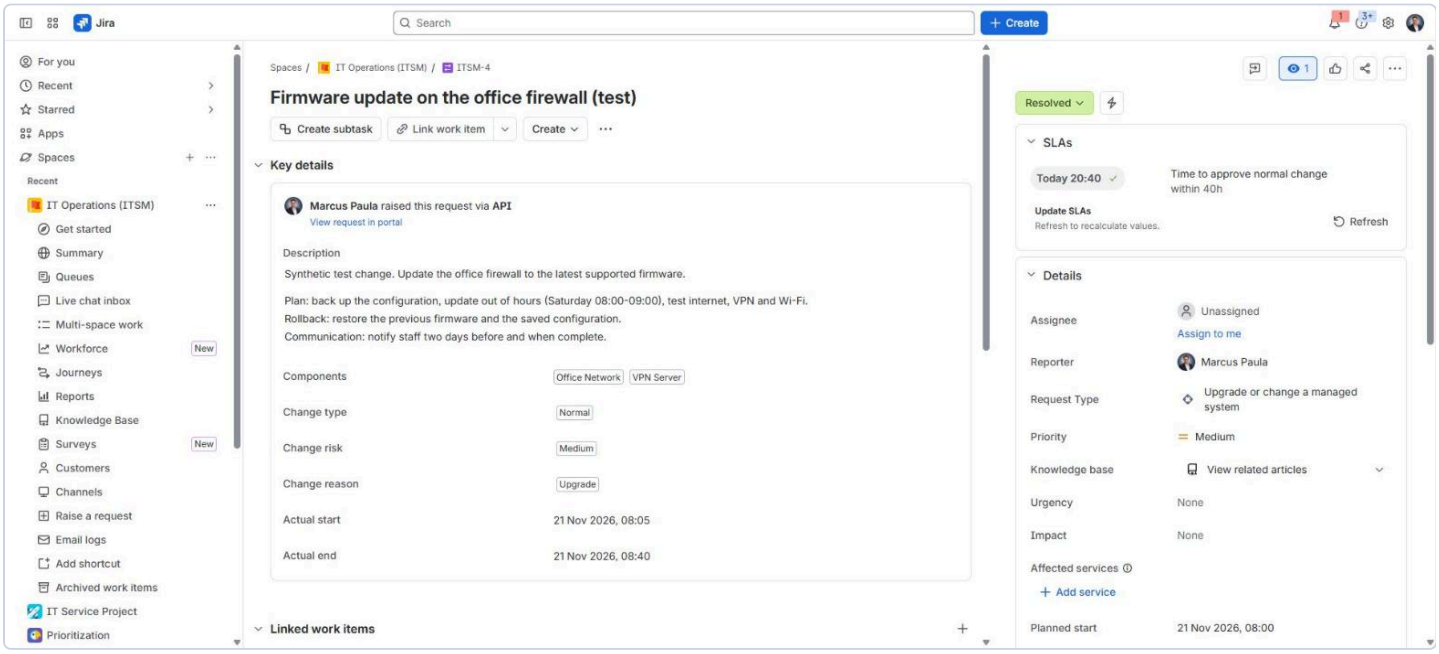
The process I follow

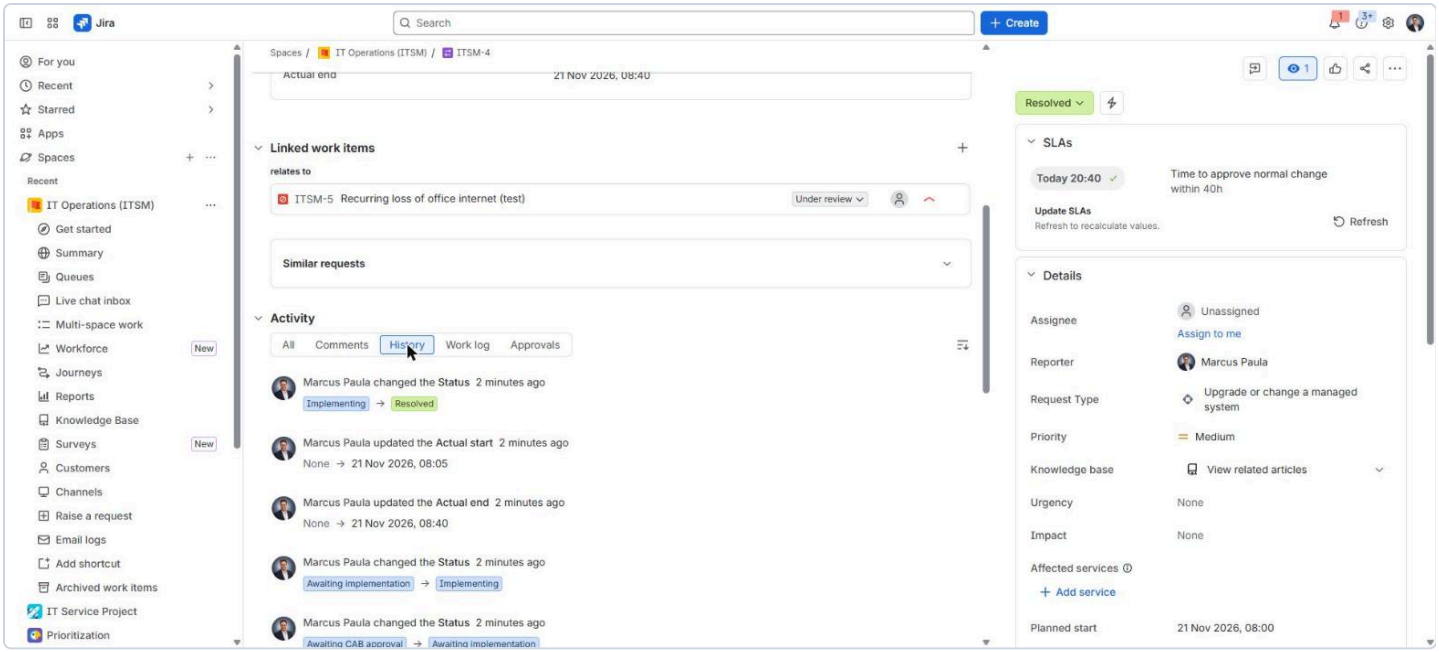


1. Export the report of missing updates and open alerts.
2. Sort by severity and by the number of devices affected.
3. Check that the devices are checking in. A device that does not check in cannot get the update.
4. If there is an active threat, raise an urgent change and inform the security owner.
5. Otherwise, raise a normal change with a plan, a rollback and a communication step.
6. Get approval.
7. Deploy to a small pilot group first.
8. Check the pilot devices work.
9. Roll out to all devices.
10. Run the report again. Close the alerts that are fixed. Record any exception with an owner and a date.

Proof

There is no Defender or patch tool screenshot in the lab. The lab change `ITSM-4` is the model I use for a patch change: plan, rollback, communication and approval before any work.





- Project 01: post-image checklist with Windows and vendor updates
- Read-only script: Get-EndpointReadiness.ps1, includes the Windows Update service state
- Admin guide: change workflow and CAB
- Build log, Step 12

Tools

- Windows Update, Jira Service Management change management
- Familiar: same process, different system. Microsoft Defender, Intune update rings, Jamf.

Lab work uses synthetic data. No employer data is published.

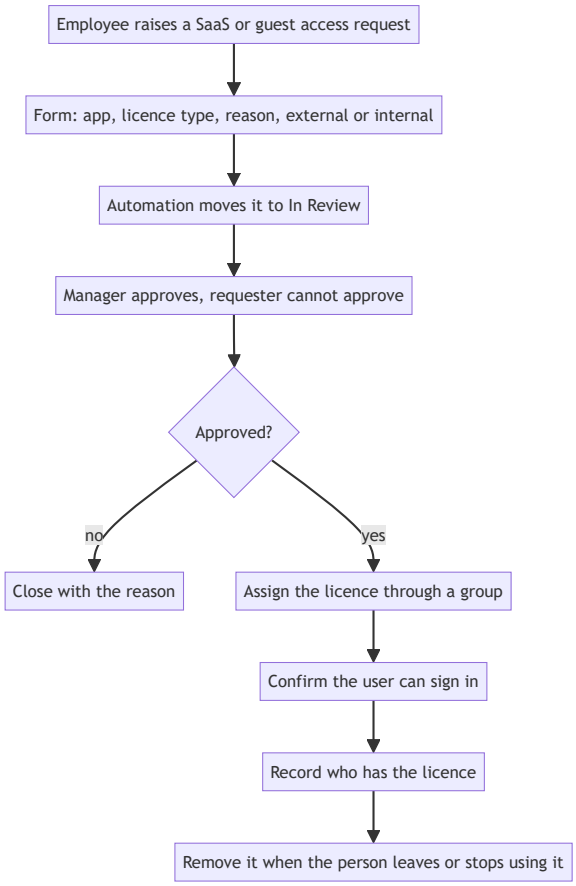
SaaS licence and external access request

The situation: An employee asks for a paid app licence. Another asks to give an outside partner access to a work tool. Both cost money or open data.

What I did in real work

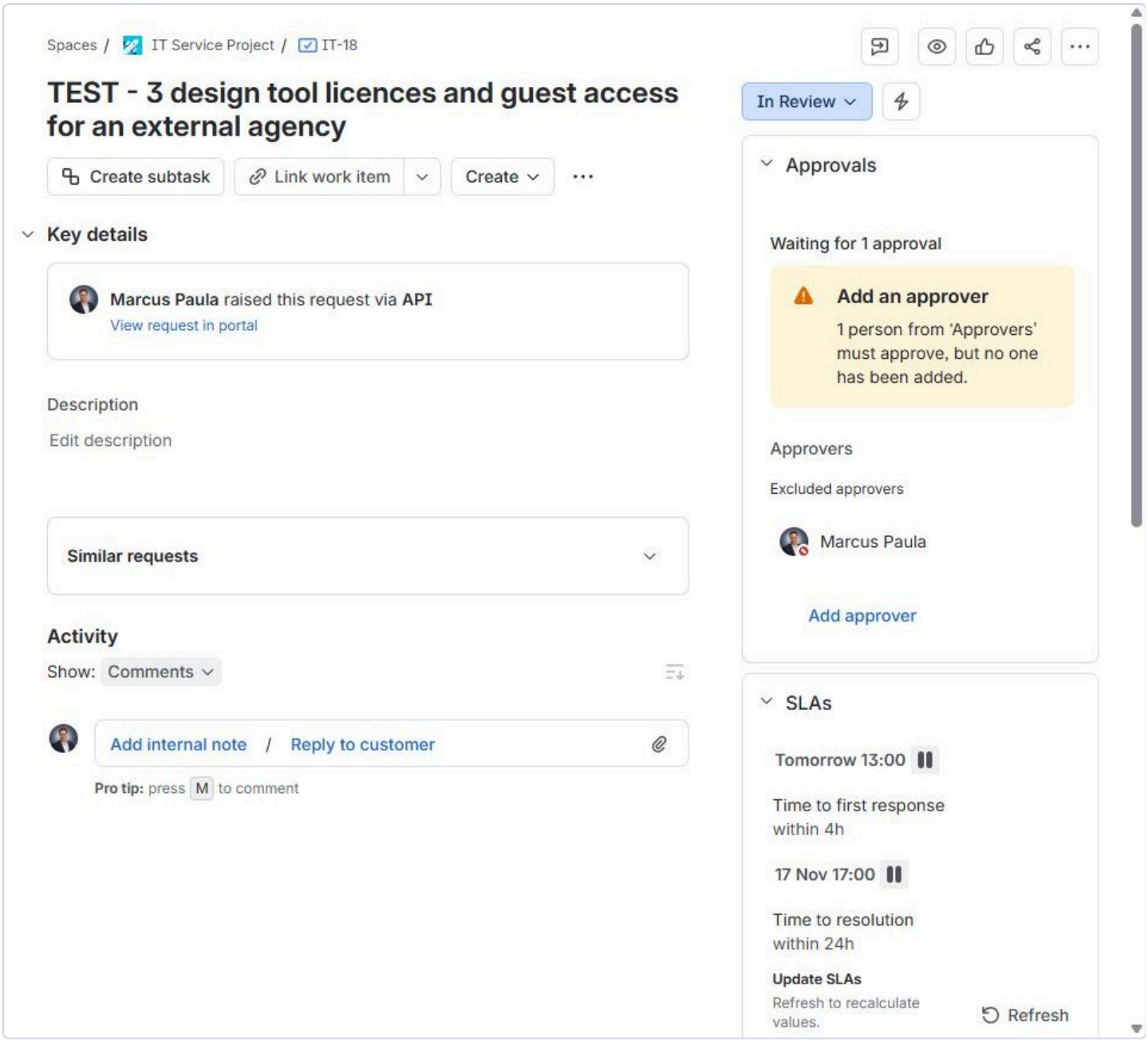
At TikTok I assigned Microsoft 365 licences and added people to security groups based on their role. I did not administer other SaaS apps there.

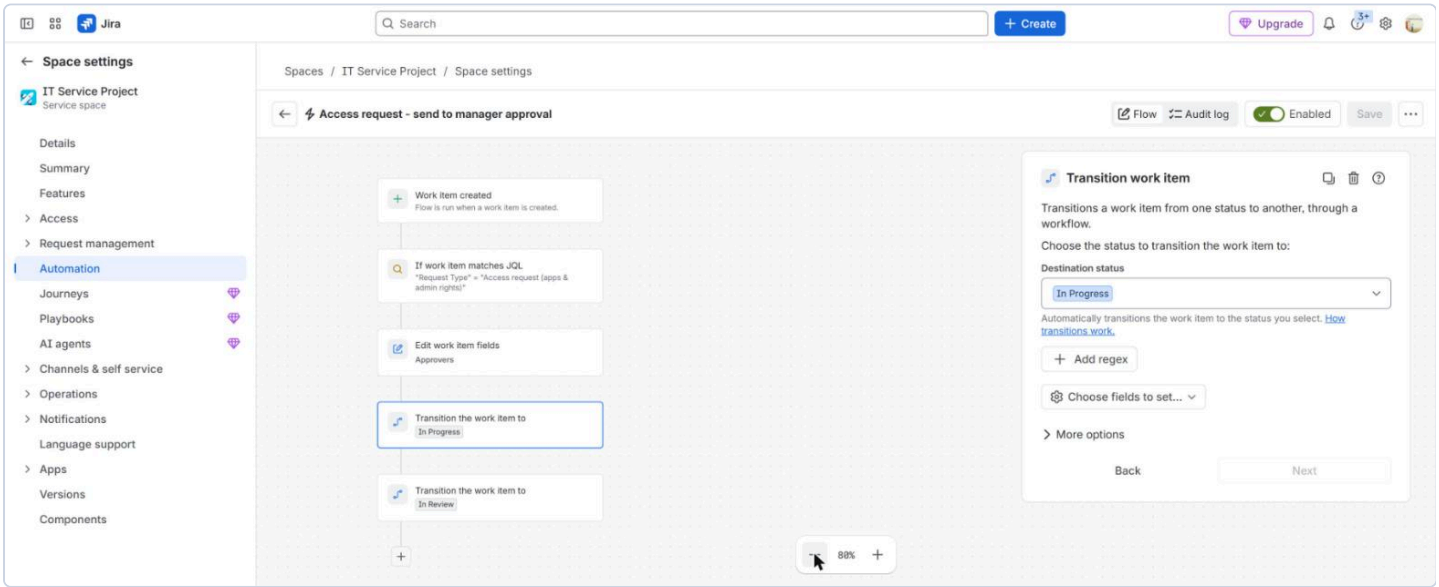
The process I follow

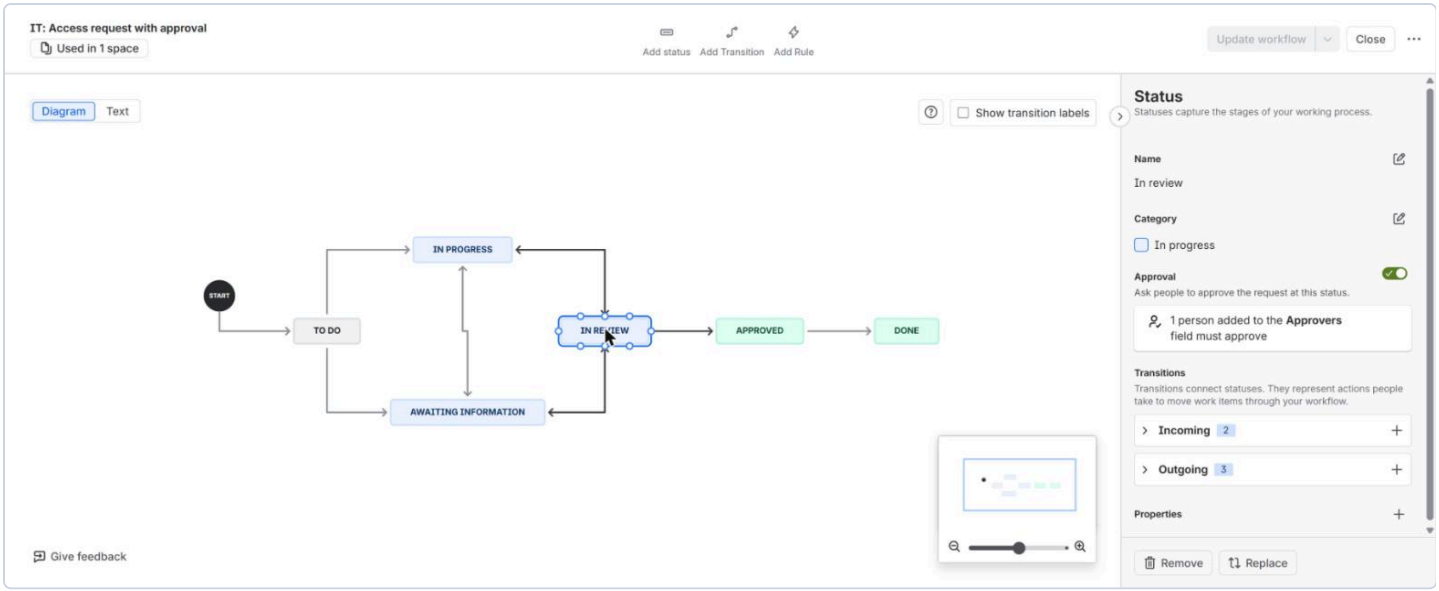


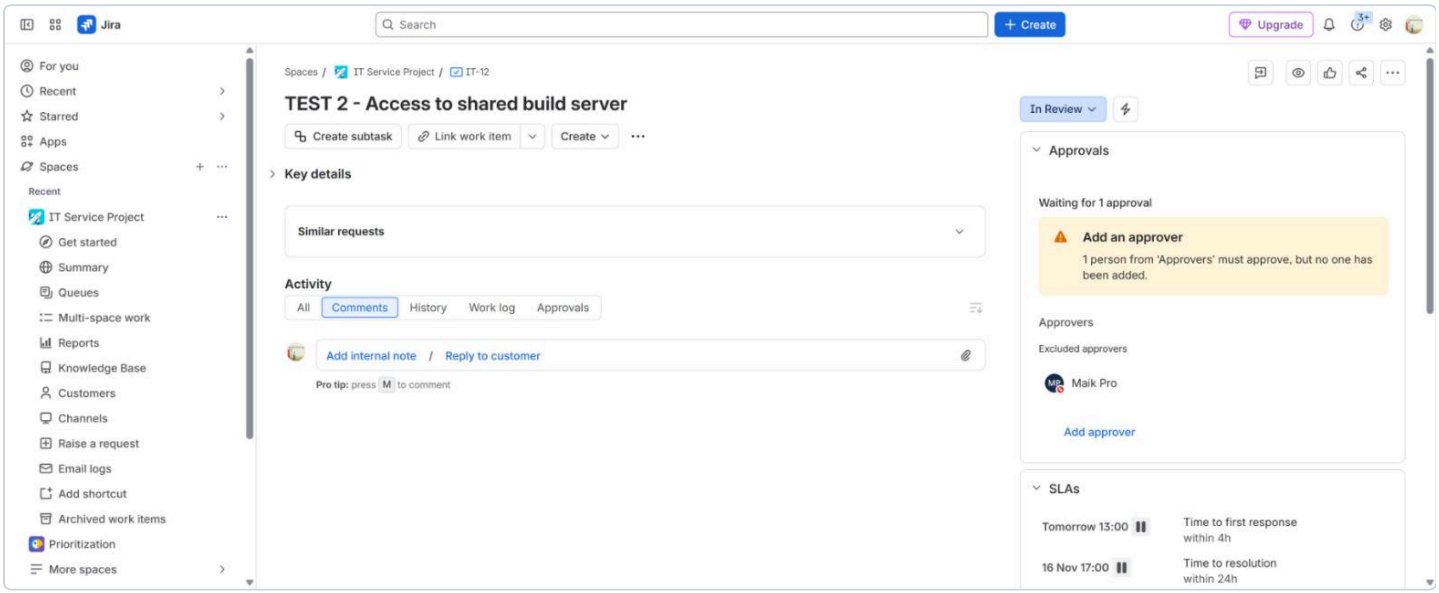
1. The employee raises the request in the portal.
2. The form asks for the app, the licence type, the reason and whether a guest is external.
3. Automation moves the request to In Review.
4. The manager approves or rejects it. The requester cannot approve their own request.
5. If it is rejected, close the request with the reason.
6. If it is approved, assign the licence through a group.
7. Confirm the user can sign in.
8. Record who has the licence.
9. Remove the licence when the person leaves or stops using the app.

Proof


Spaces / IT Service Project / IT-18
TEST - 3 design tool licences and guest access for an external agency
In Review
Create subtask
Link work item
Create
Key details
Marcus Paula raised this request via API
View request in portal
Description
Edit description
Similar requests
Activity
Show: Comments
Add internal note / Reply to customer
Pro tip: press M to comment
Approvals
Waiting for 1 approval
Add an approver
1 person from 'Approvers' must approve, but no one has been added.
Approvers
Excluded approvers
Marcus Paula
Add approver
SLAs
Tomorrow 13:00
Time to first response within 4h
17 Nov 17:00
Time to resolution within 24h
Update SLAs
Refresh to recalculate values.
Refresh


Jira
Search
Create
Upgrade
Space settings
IT Service Project
Service space
Details
Summary
Features
Access
Request management
Automation
Journeys
Playbooks
AI agents
Channels & self service
Operations
Notifications
Language support
Apps
Versions
Components
Spaces / IT Service Project / Space settings
Access request - send to manager approval
Flow
Audit log
Enabled
Save
Work item created
If work item matches JQL
Edit work item fields
Approvers
Transition the work item to
In Progress
Transition the work item to
In Review
Transition work item
Transitions a work item from one status to another, through a workflow.
Choose the status to transition the work item to:
Destination status
In Progress
Automatically transitions the work item to the status you select. How transitions work.
Add regex
Choose fields to set...
More options
Back
Next
80%





- Build log, Step 15: SaaS licence or external guest access, test IT-18
- Build log, Step 8: workflow and approvals
- Build log, Step 9: the failed run and the fix
- Project 09: group-based licensing for joiners and leavers

Tools

- Microsoft 365 licences, Entra ID groups, Jira Service Management approvals
- Familiar: same process, different system. At TikTok we used Lark. Similar apps such as Teams, Slack, Adobe, Autodesk and Miro follow the same process.

Lab work uses synthetic data. No employer data is published.

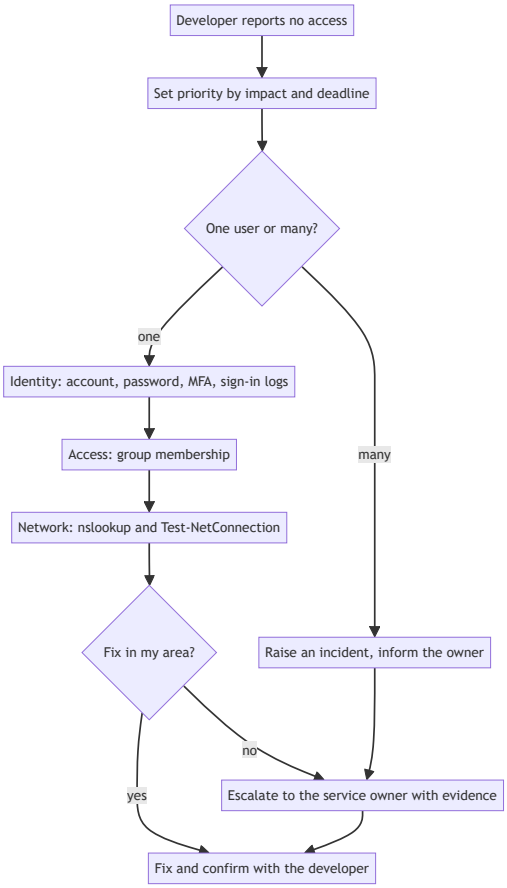
Developer cannot access a critical service

The situation: A developer cannot reach the build server or a remote workstation on the day of a release.

What I did in real work

At TikTok I supported Entra ID sign-in and group problems, and I owned L2 and L3 escalations in Jira. When the service belonged to another team, I worked this with the owning team and escalated with evidence.

The process I follow



1. Ask what fails, the exact error and since when.
2. Set the priority by impact and by the release deadline.
3. Check if one user or many users are affected. Many users means an incident for the service owner.
4. Check the account: enabled, password, MFA and the sign-in logs.
5. Check group membership for the service.
6. Check the network path: `nslookup build.example.com` and `Test-NetConnection build.example.com -Port 443`
7. If the fix is in my area, fix it.
8. If not, escalate to the service owner with the results of steps 4 to 6.
9. Confirm with the developer that access works.
10. Write the fix in the ticket.

Proof

Process only, no screenshot in the lab for this tool. The closest proof:

- Troubleshooting guide: sign-in, MFA and access problems
- Read-only script for account state, groups and MFA methods: Get-UserAccessSnapshot.ps1
- When the developer needs new access, it goes through approval: IT-12 waiting for approval
- Build log, Step 12: incident handling
- Agent runbook: triage

Tools

- Entra ID sign-in logs, `nslookup`, `Test-NetConnection`, Jira Service Management
- Familiar: same process, different system. Jenkins and Parsec access, SSO.

Lab work uses synthetic data. No employer data is published.

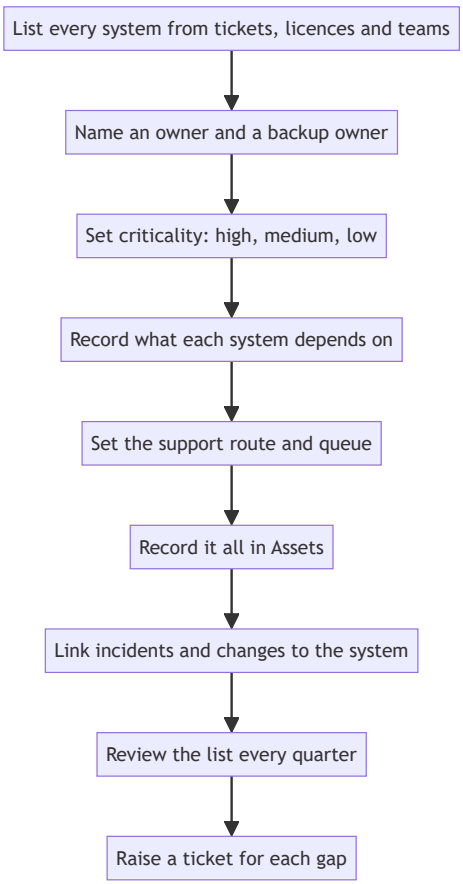
Systems inventory and ownership

The situation: The company runs about 30 systems. Nobody has one list that says who owns each system, how critical it is and who to call when it fails.

What I did in real work

At TikTok I helped design a scan-based asset audit. It covered more than 5,000 assets and reached 99% visibility. That was hardware. I did not build a register of systems and services there. For those, I worked with the owning teams.

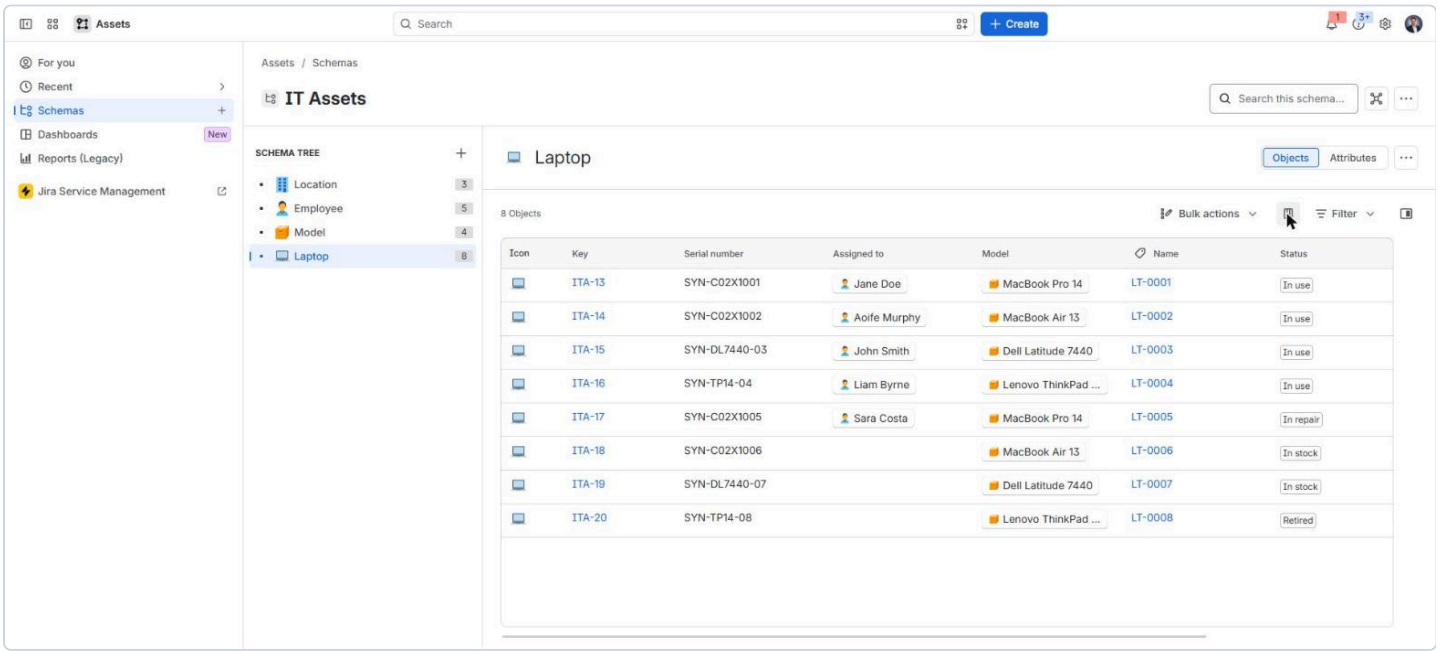
The process I follow

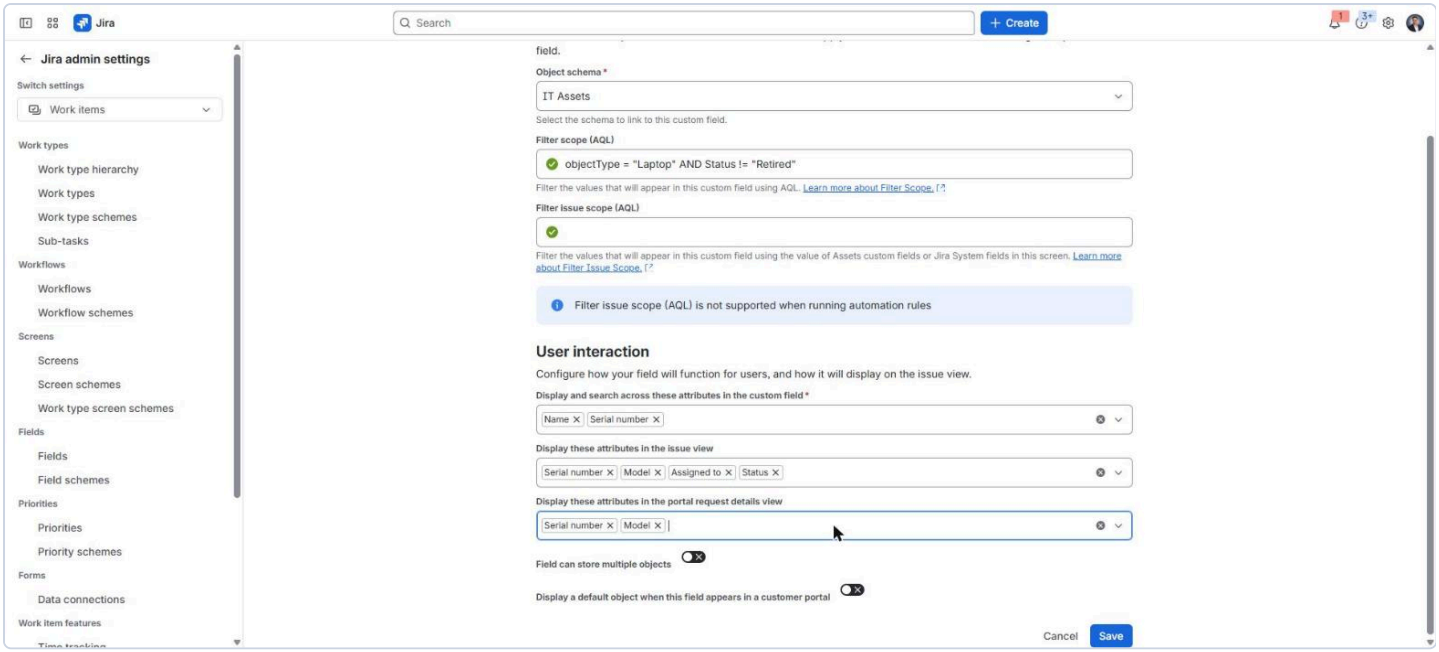


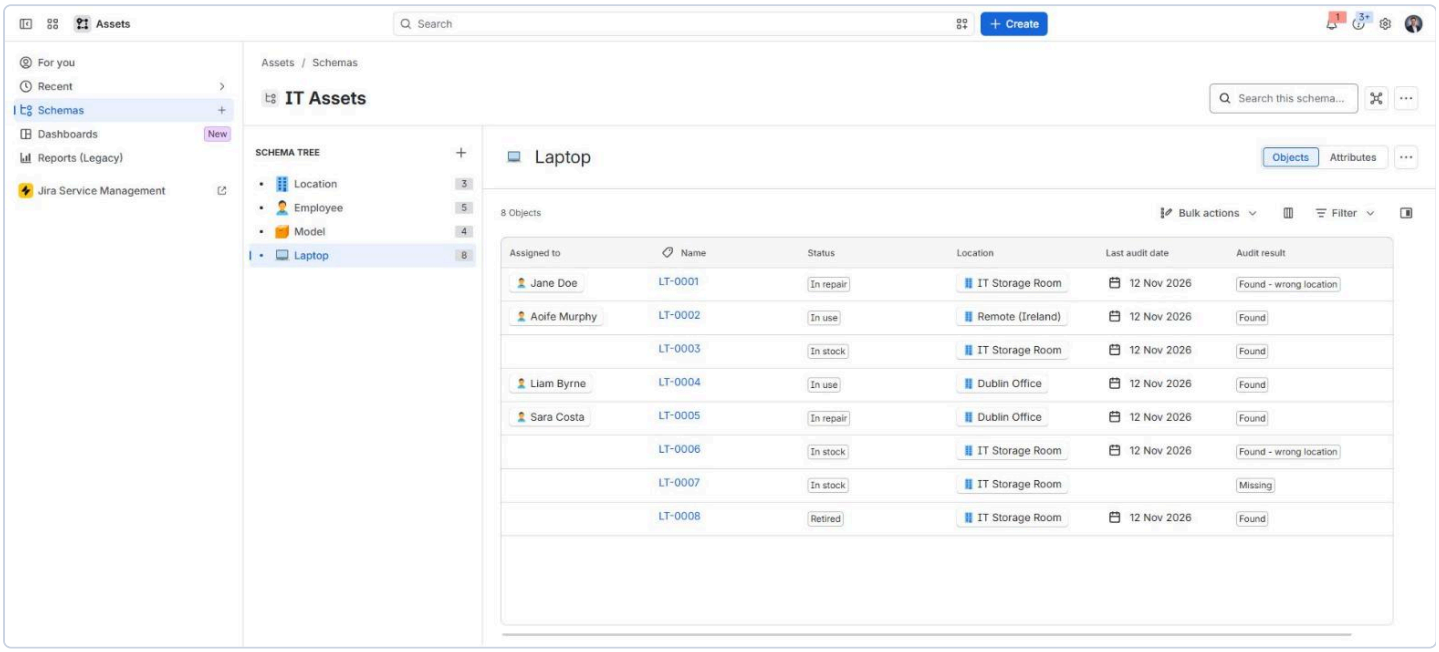
1. Build the list of systems from tickets, licences, invoices and team leads.
2. Name an owner and a backup owner for each system.
3. Set the criticality: what stops if the system fails, and for how long the business can wait.
4. Record what each system depends on, for example the network, identity or a supplier.
5. Set the support route: which queue and which team.
6. Record the systems in Assets, with owner and criticality as attributes.
7. Link incidents and changes to the system they affect.
8. Review the list every quarter with the owners.
9. Raise a ticket for each gap, for example a system with no owner.

Proof

The lab has a hardware register with owners and locations. A systems register with owner and criticality is design only, not built.







- Configuration: Assets schema
- Admin guide: Assets schema and AQL
- Procedure: scan-based asset audit
- Audit reconcile tool and its tests
- Project 02: the real asset audit
- Build log, Step 13: audit and return

Tools

- Jira Service Management Assets, AQL, Python, CSV exports

Lab work uses synthetic data. No employer data is published.

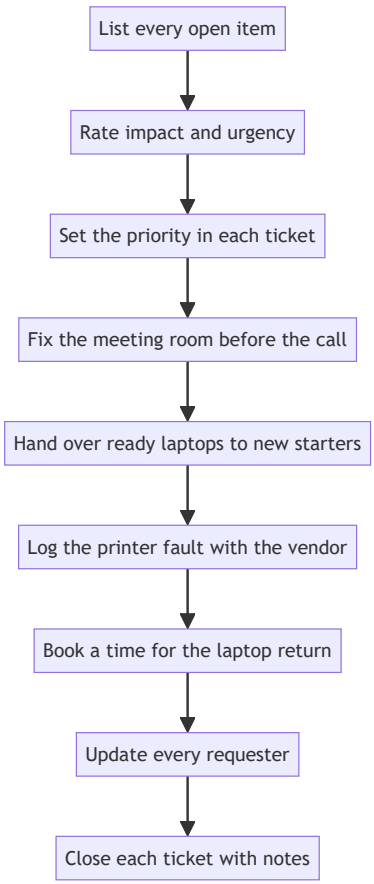
On-site prioritisation

The situation: It is Monday morning. Two new starters arrive at 10:00, a meeting room fails before a leadership call, a printer is down and a leaver is returning a laptop.

What I did in real work

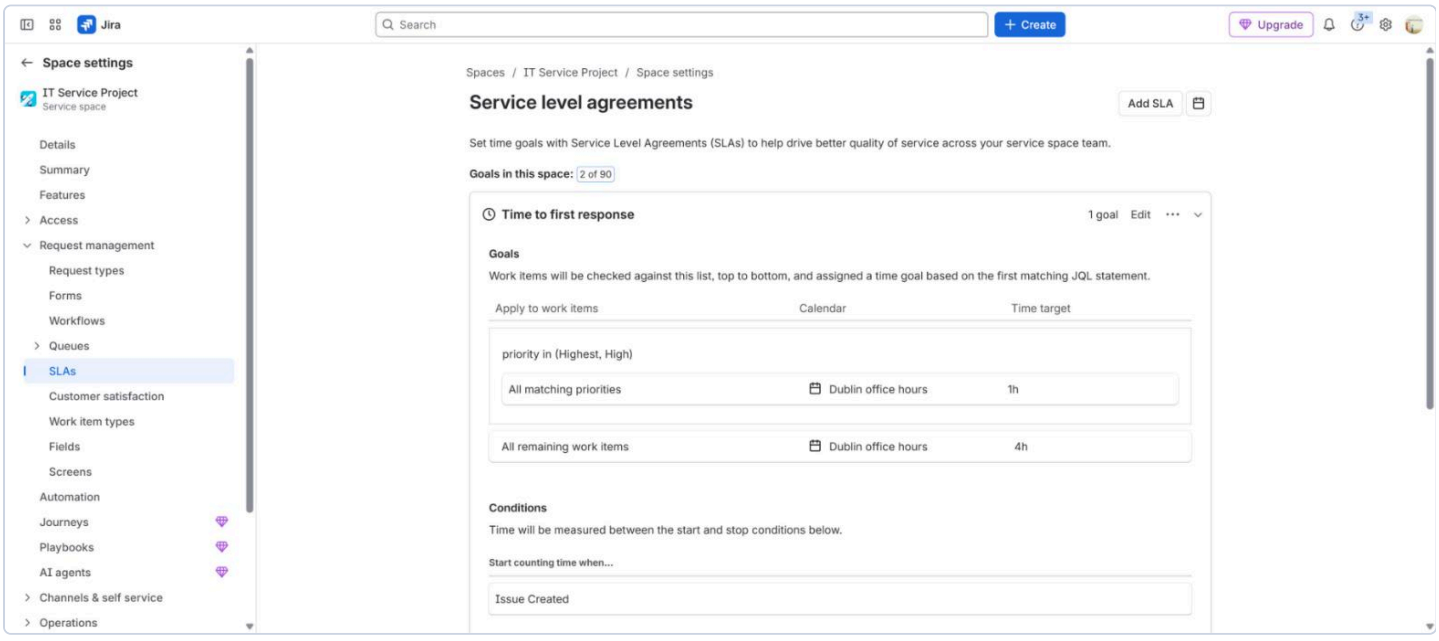
At TikTok I set priorities on tickets every day and worked to SLAs. I looked after 19 meeting rooms on Lark and Zoom. I worked on the asset return process in three sites, and I handled logistics with couriers and vendors.

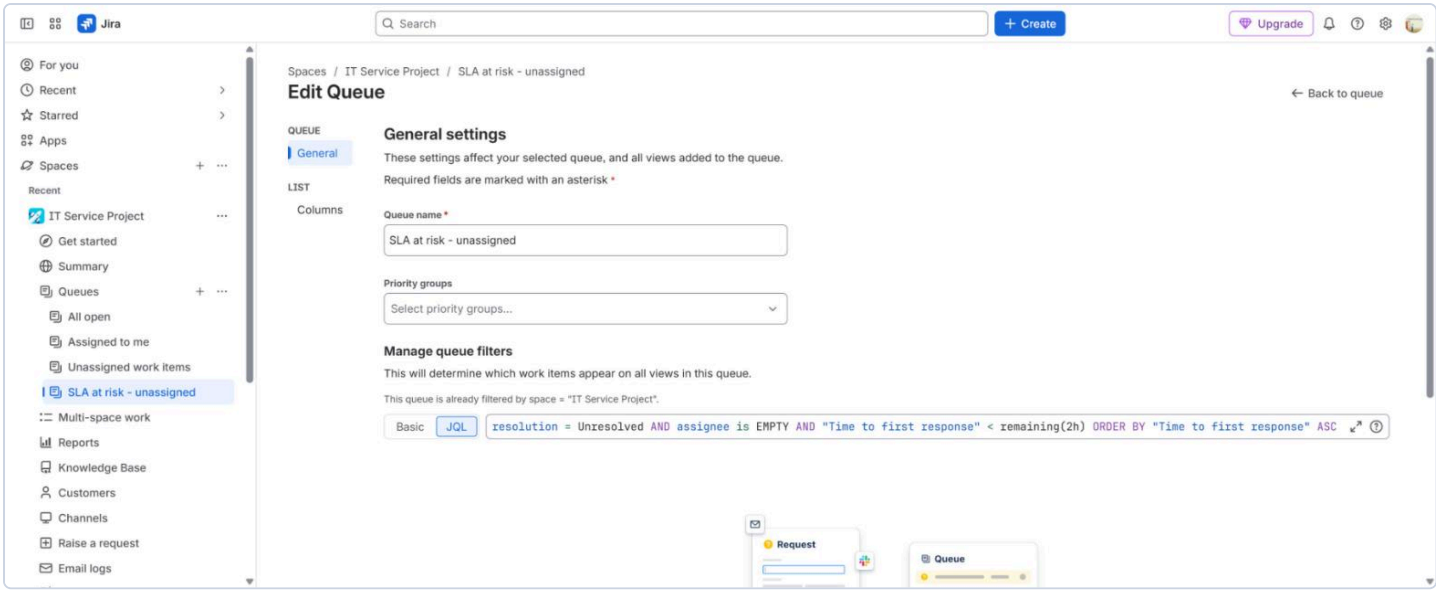
The process I follow

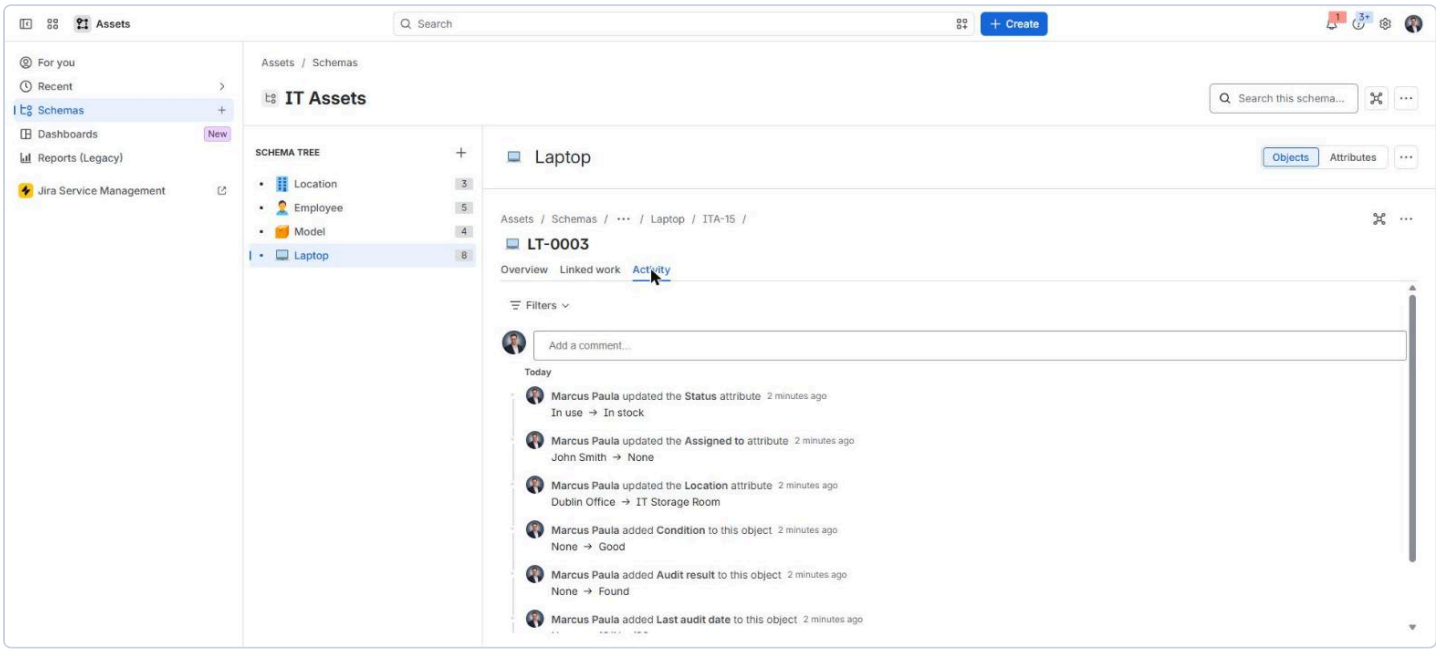


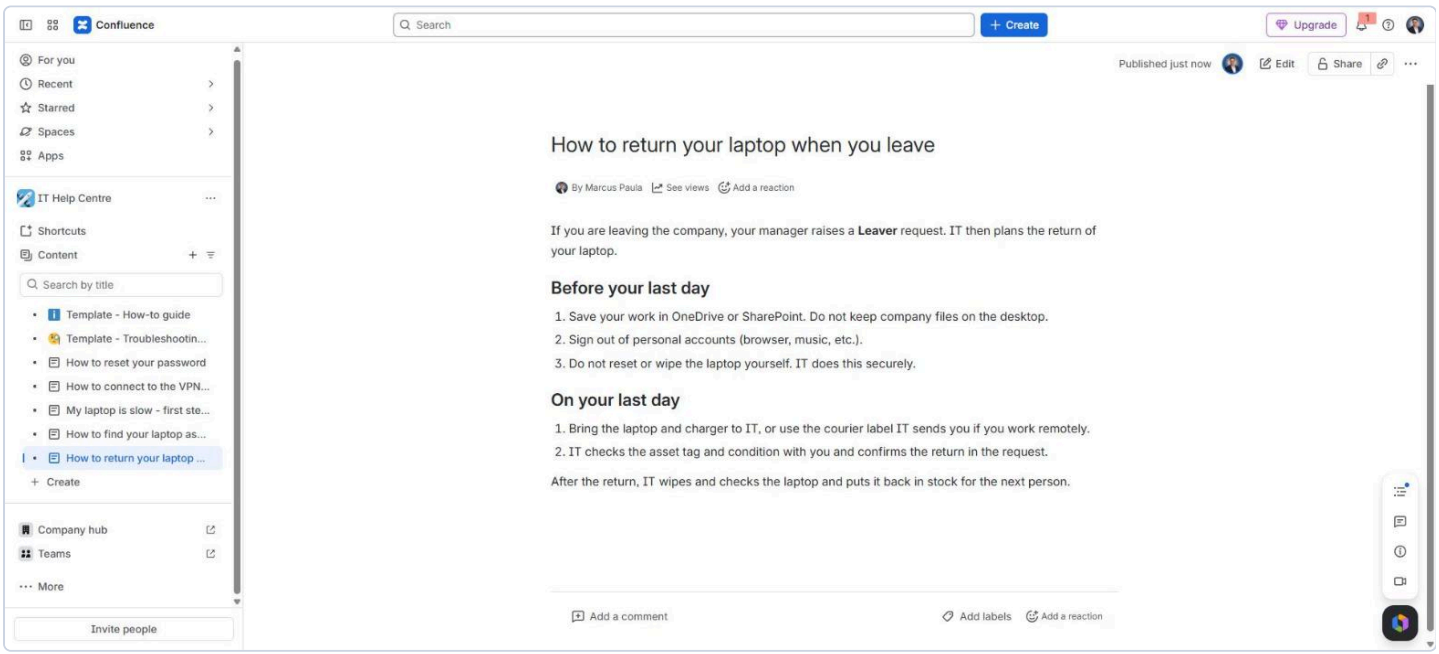
1. List every open item in one place.
2. For each item, ask: how many people are blocked, and is there a fixed time?
3. Set the priority in each ticket.
4. Fix the meeting room first. The call has a fixed time and many people.
5. Hand over the new starter laptops. Accounts and laptops were prepared before the day.
6. Log the printer fault with the vendor and tell staff which printer to use now.
7. Book a time for the laptop return. Check the asset tag and the condition when it arrives.
8. Update every requester on what happens next and when.
9. Close each ticket with notes.

Proof









- Project 05: meeting room readiness, 19 rooms and the room checklist
- Project 04: asset return in three sites
- Agent runbook: triage
- Agent runbook: laptop return on a Leaver request
- Build log, Step 4: SLAs

Tools

- Jira Service Management, Lark, Zoom, courier and vendor portals

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Lab work uses synthetic data. No employer data is published.

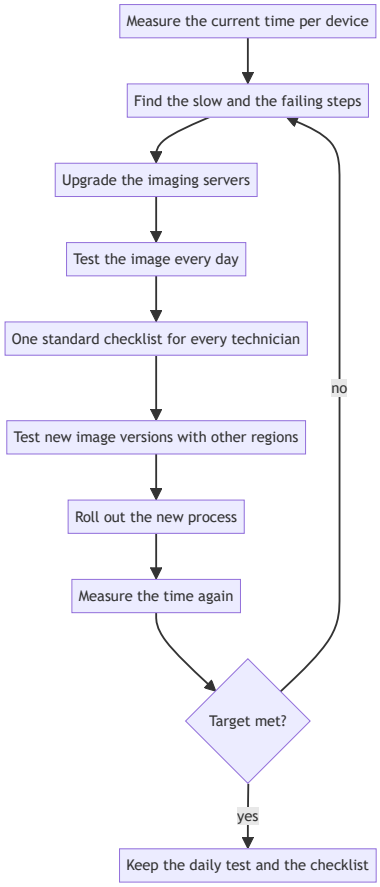
Process improvement: imaging from 60 to 15 minutes

The situation: Preparing one laptop takes about an hour. Results are different from one technician to the next, and laptops wait in a queue.

What I did in real work

At TikTok I cut imaging time from about 60 minutes to 15 minutes per device. We deployed the images with MCU. I upgraded the imaging servers, set up a daily image test, introduced a standard checklist, and tested image versions with colleagues in other regions.

The process I follow



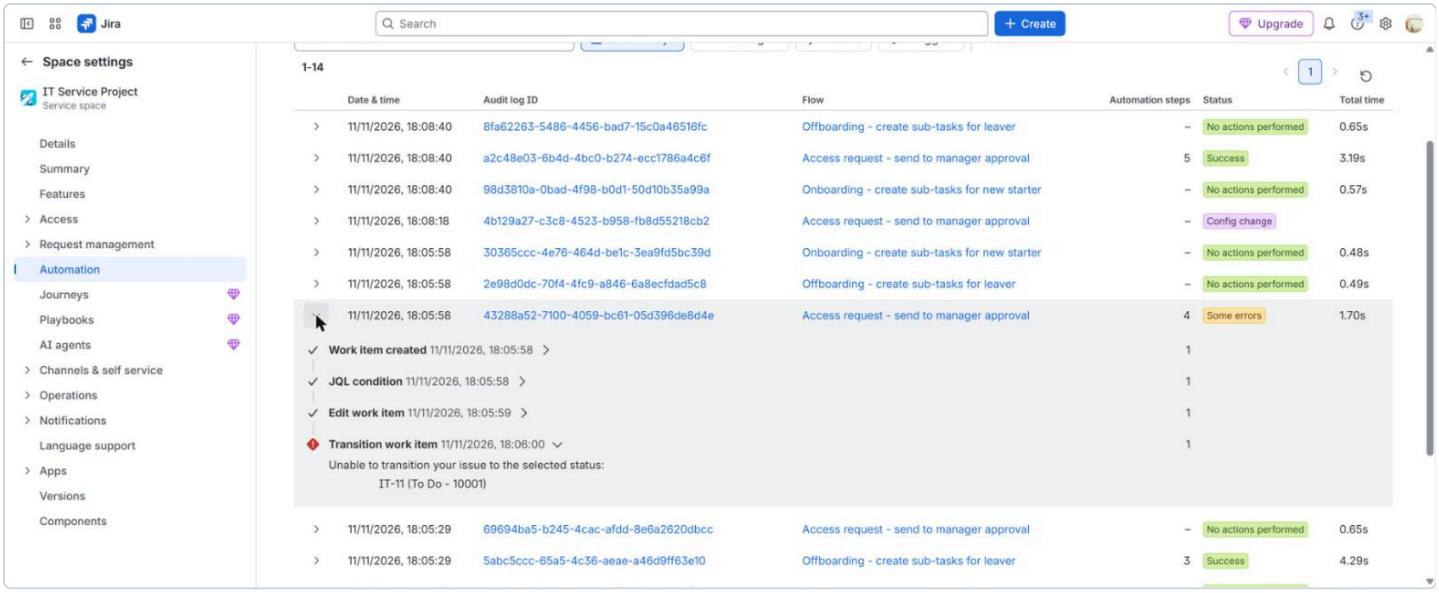
1. Measure how long one device takes today, from start to ready.
2. Find the steps that are slow and the steps that fail.
3. Upgrade the imaging servers.
4. Test the image every day on a test device, so a bad image is found before it reaches users.
5. Write one standard checklist that every technician follows.
6. Test each new image version with colleagues in other regions before it goes live.
7. Roll out the new process.
8. Measure the time per device again.
9. If the target is not met, go back to step 2.
10. Keep the daily test and the checklist as normal work.

Proof

Process only, no screenshot in the lab for this tool. The closest proof:

- Project 01: imaging improvement
- Project 01: post-image validation checklist
- Read-only check after imaging: Get-EndpointReadiness.ps1

The same habit in the lab: test, find the cause, fix, test again. A flow failed, the audit log showed a missing workflow step, and the next test passed.



- Build log, Step 9: testing and troubleshooting

Tools

- MCU for image deployment, Linux and Windows imaging servers, PowerShell
- Familiar: same process, different system. Intune, Jamf and Mosyle zero-touch enrolment.

Author: Marcus Paula
Lab work uses synthetic data. No employer data is published.

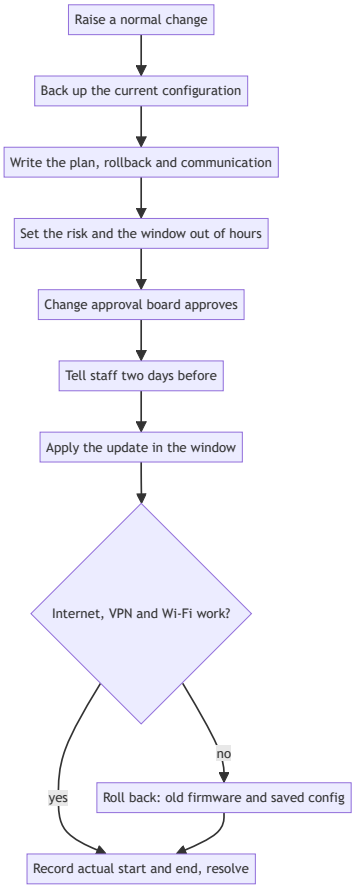
Firewall management

The situation: The office firewall needs a firmware update. A mistake can stop internet and VPN for everyone in the office.

What I did in real work

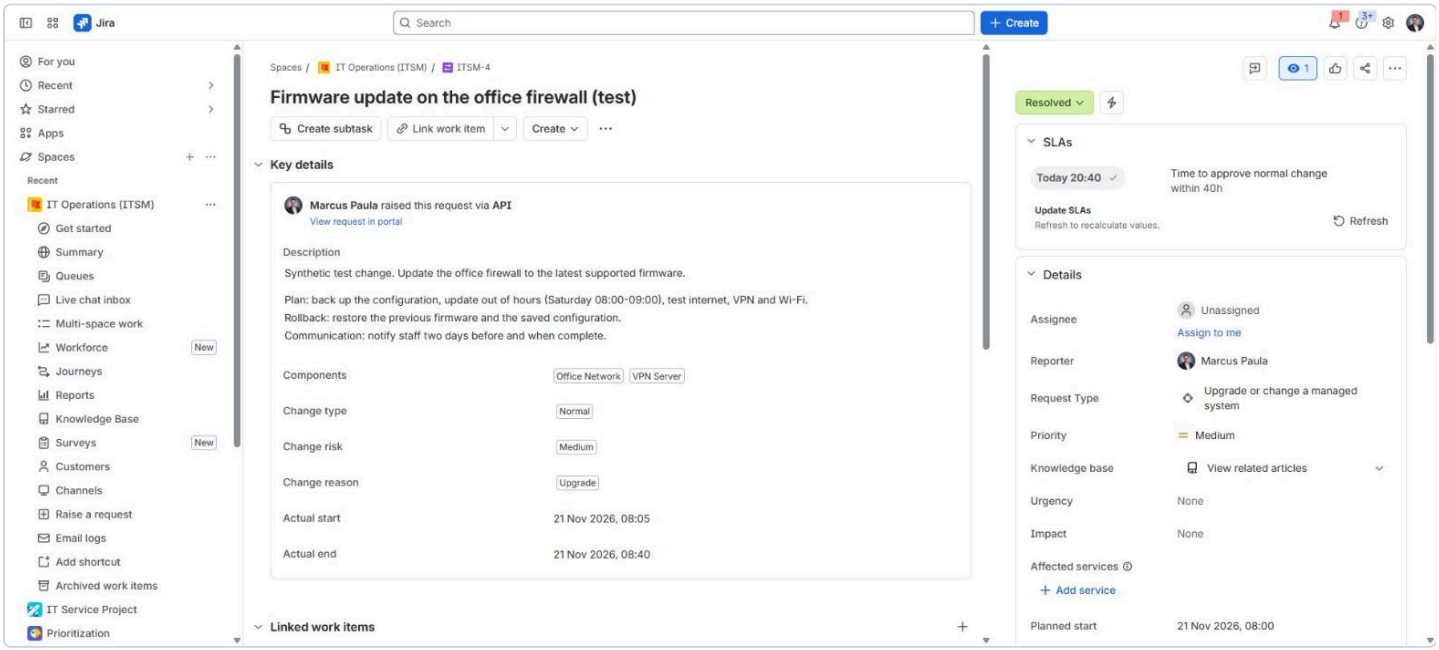
At TikTok the firewall and VPN were owned by the network team. I did the on-site troubleshooting with `nslookup` and `Test-NetConnection`, and I escalated to the network team with evidence.

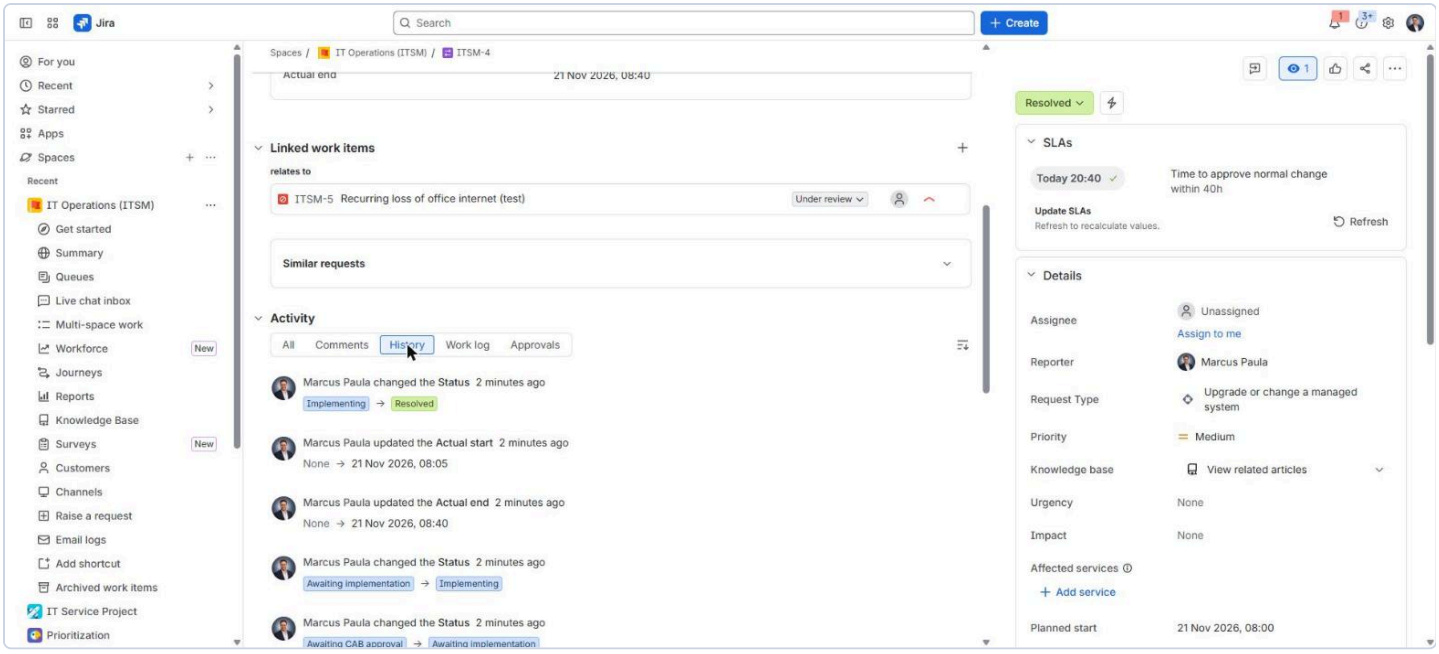
The process I follow



1. Raise a normal change for the firmware update.
2. Back up the current firewall configuration.
3. Write the plan, the rollback and the communication.
4. Set the risk and pick a window out of hours.
5. Get approval from the change approval board.
6. Tell staff two days before the window.
7. Apply the update in the window.
8. Test internet, VPN and Wi-Fi.
9. If a test fails, roll back to the old firmware and the saved configuration.
10. Record the actual start and end times, tell staff it is done and resolve the change.

Proof





- Build log, Step 12: change ITSM-4
- Admin guide: change workflow and CAB
- Configuration: change workflow and CAB approval
- Related outage and its problem record: scenario 08

Tools

- Jira Service Management change management, `nslookup`, `Test-NetConnection`
- Familiar: same process, different system. FortiGate rules, VPN, firmware and configuration backups.

Lab work uses synthetic data. No employer data is published.

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